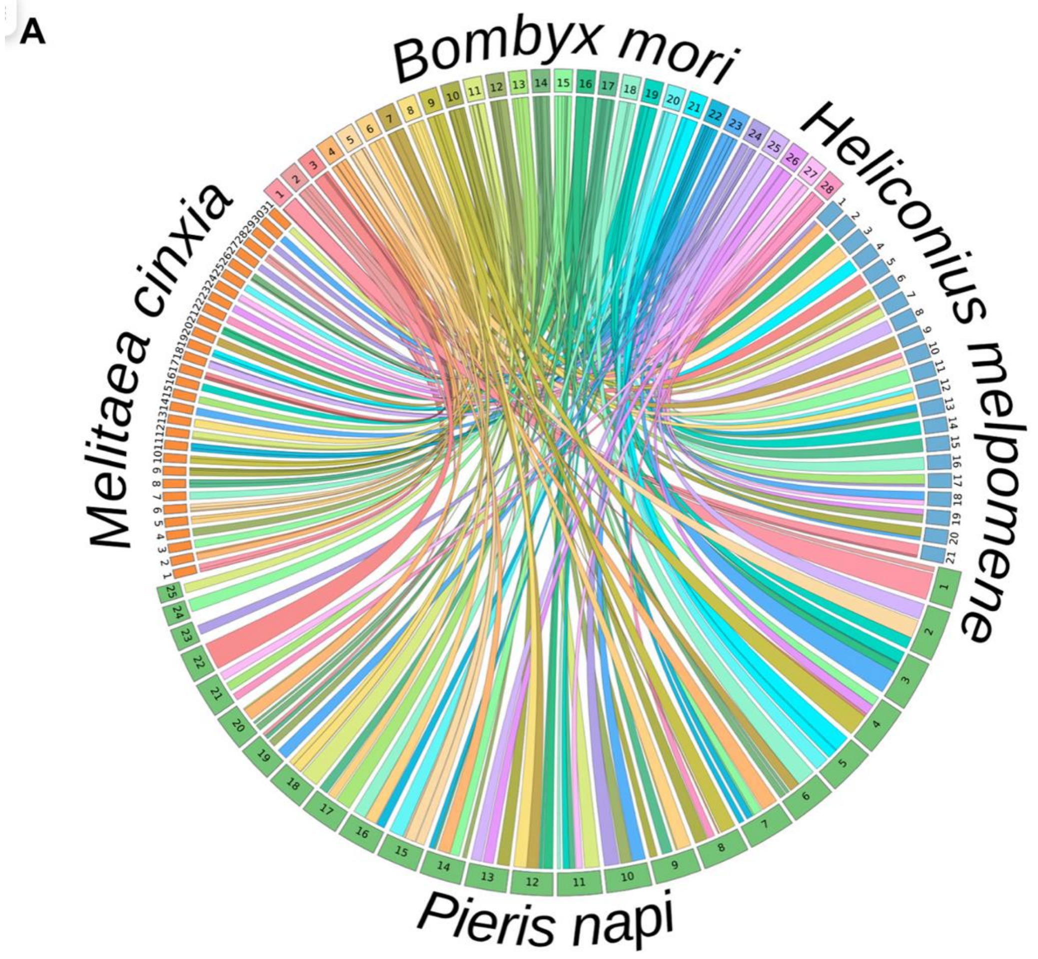
Comparative genomics using reference genomes

Introduction to
Biodiversity Genomics
Mendoza, Argentina
2025

Comparative genomics

- Genome structure
- Genome features
  - Gene and regulatory dynamics
    - Gene family dynamics
    - Conserved non-coding elements
  - TEs and repeats
  - CG-content, codon usage and tRNA dynamics
- Heterozygosity
- Signs of selection (next session)

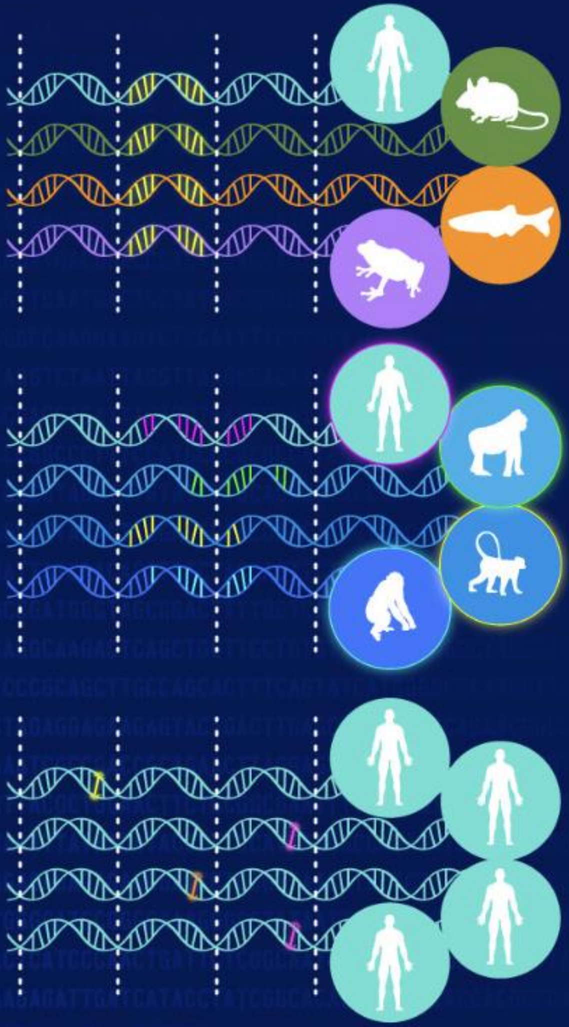


Hill et al. 2019

Comparative Genomics

NHGRI FACT SHEETS
genome.gov

Researchers choose the appropriate time-scale of evolutionary conservation for the question being addressed.



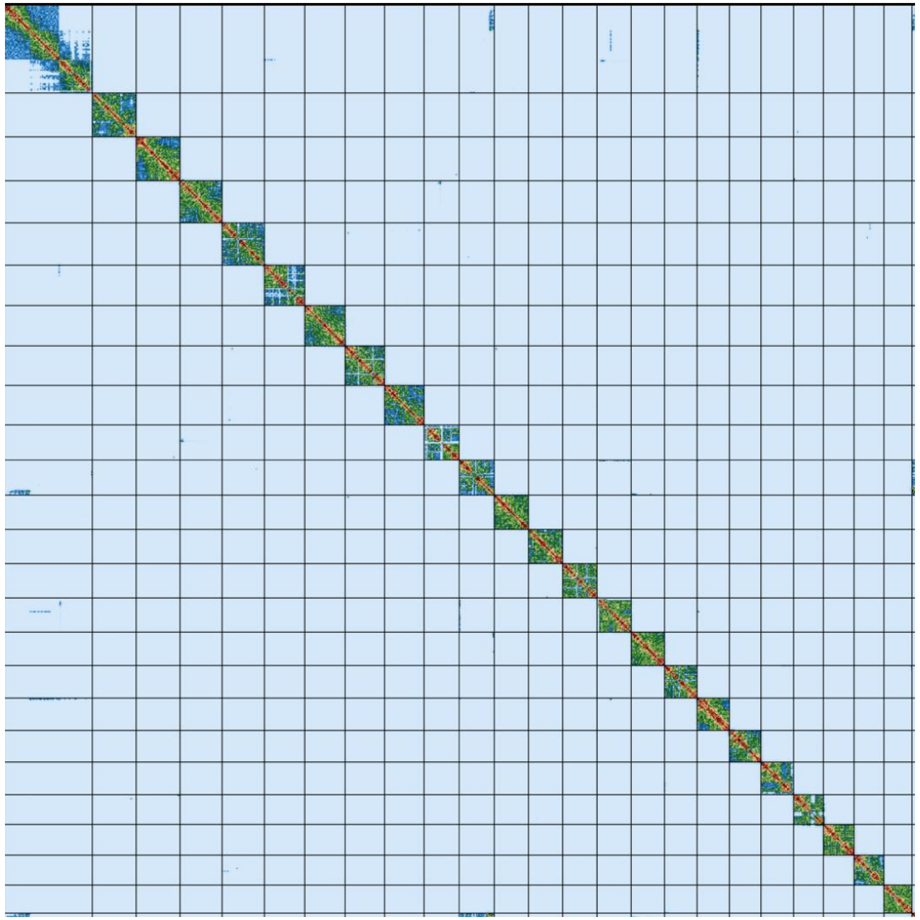
Common features of different organisms such as humans and fish are often encoded within the DNA evolutionarily conserved between them.

Looking at closely related species such as humans and chimpanzees shows which genomic elements are unique to each.

Genetic differences within one species such as our own can reveal variants with a role in disease.

NIH National Human Genome Research Institute

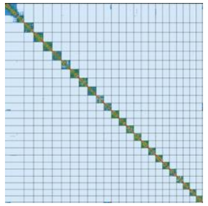
Genome structure
- size



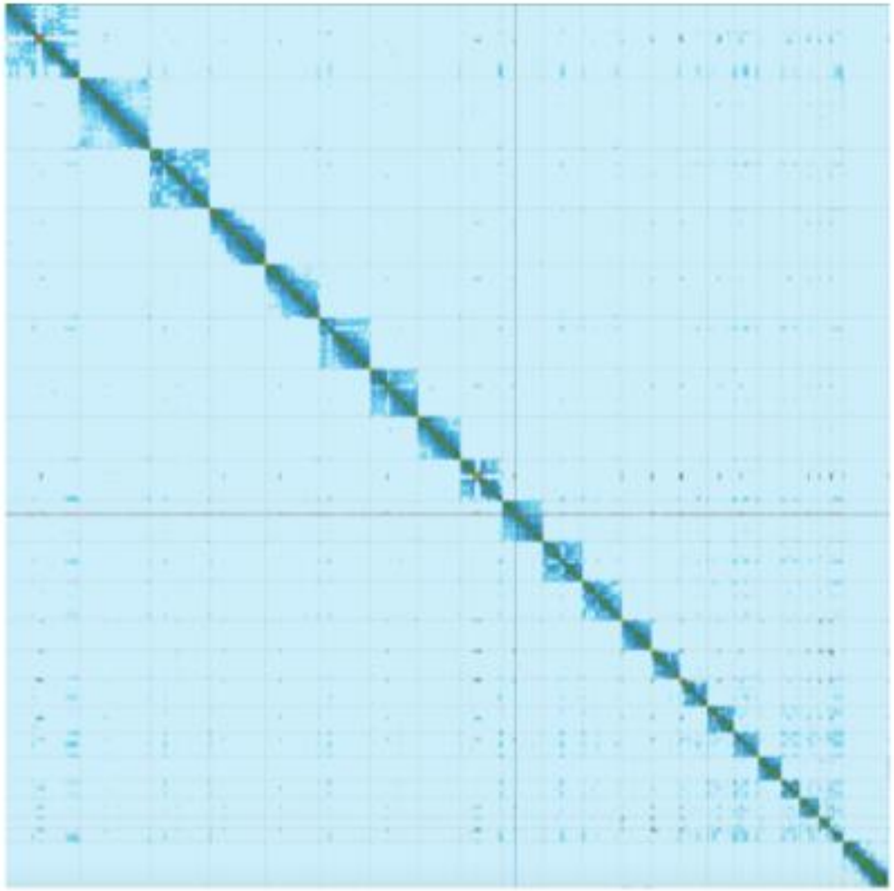
Melinaea menophilus zaneka

679 Mb

Genome structure
- size



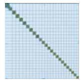
600Mb

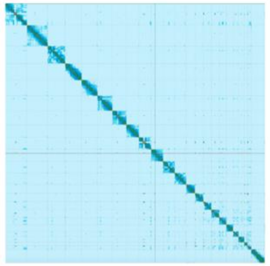


Homo sapiens, 3.1 Gb

Donaldson collection/getty images

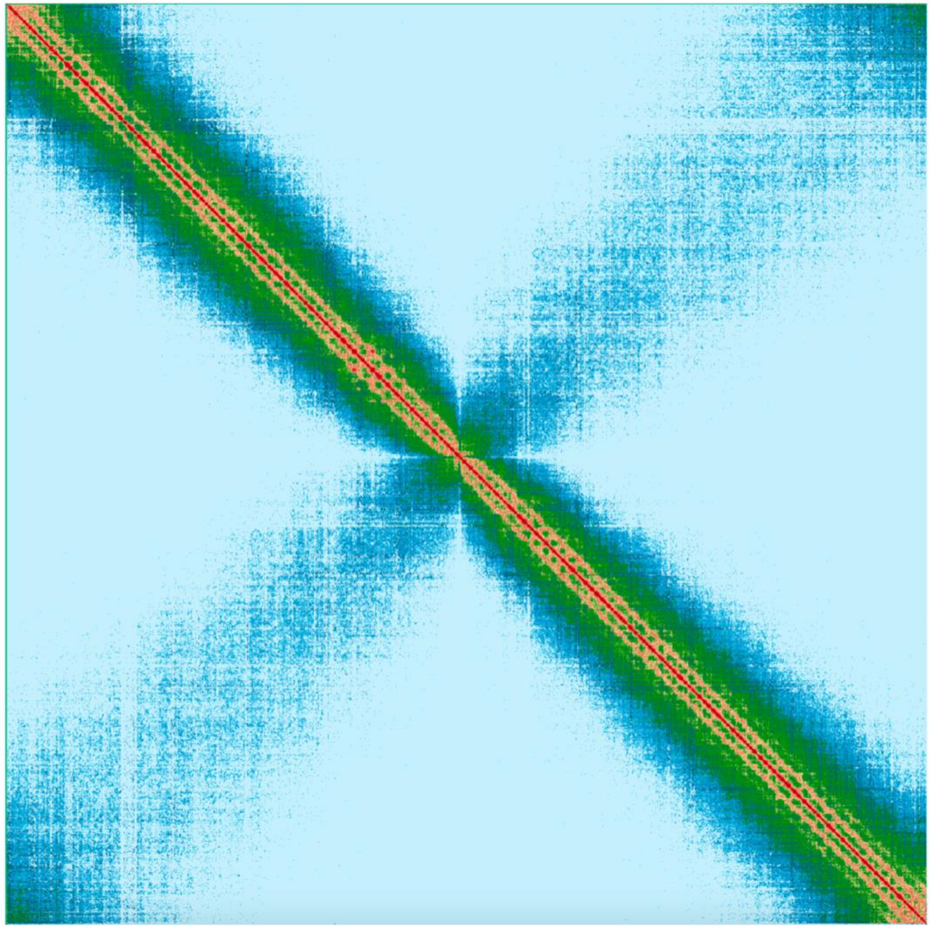
Genome structure
- size





600Mb

3.1 Gb

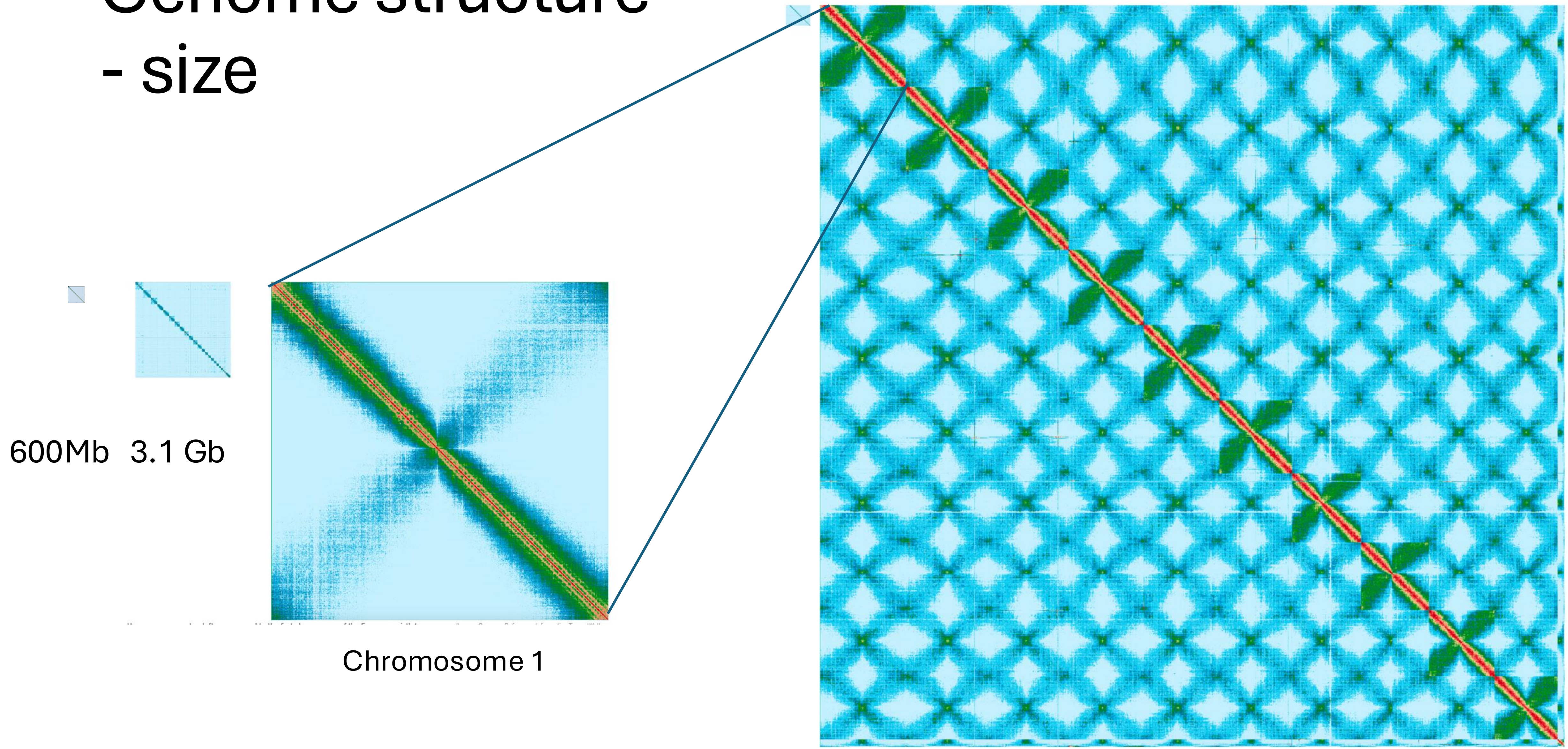


Mistletoe
Chr 1 ~11Gb

https://plantura.garden/uk/green-living/knowledge/mistletoe

Genome structure - size

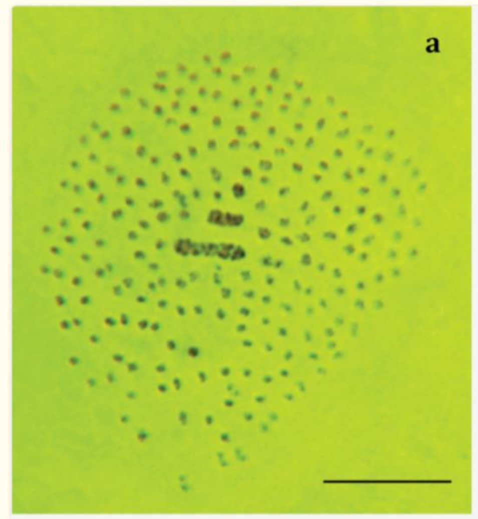
Viscum album 94.3 Gb



Human genome — again, top left — compared to the entire European mistletoe genome (Image: Genome Reference Informatics Team, Wellcome Sanger Institute)

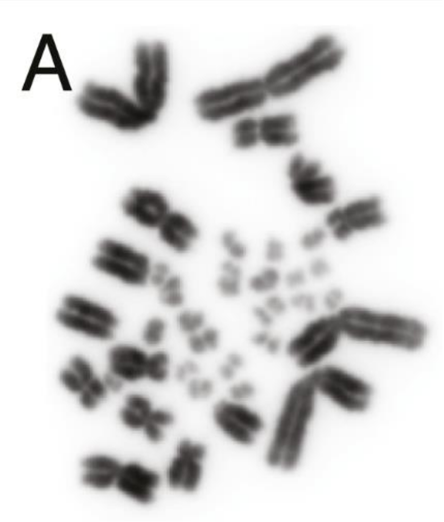
Genome structure - chromosomes

Number



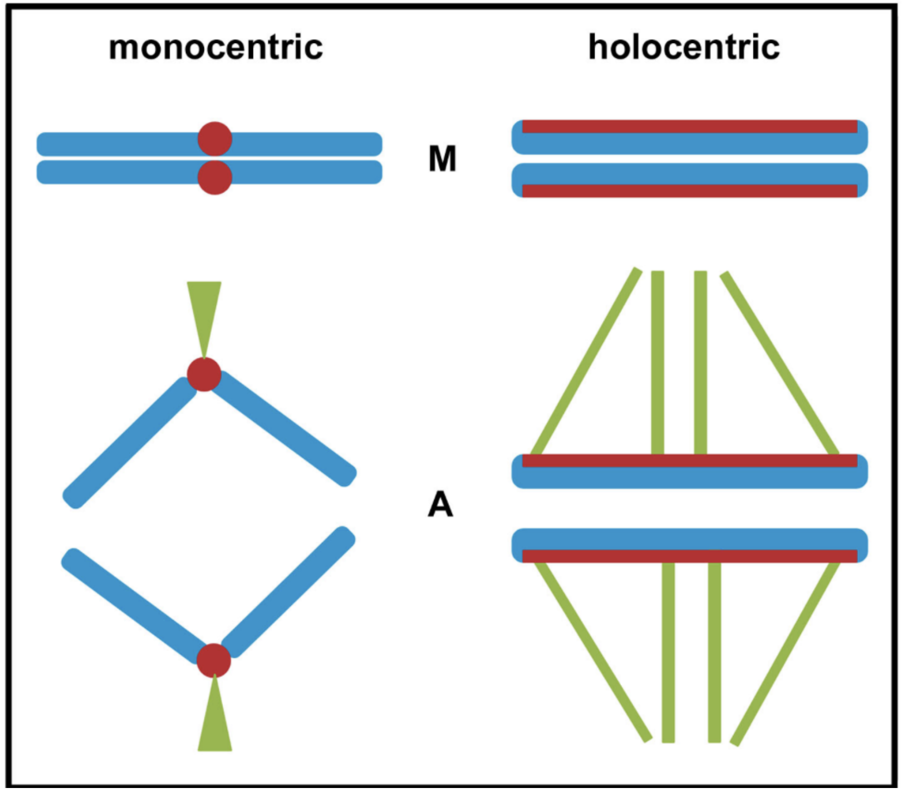
The atlas blue, n=226
Lukhtanov, 2015

Size



The bearded dragon,
Waters et al 2021

Function and organisation

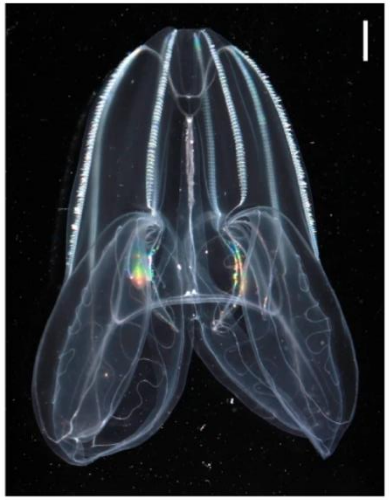


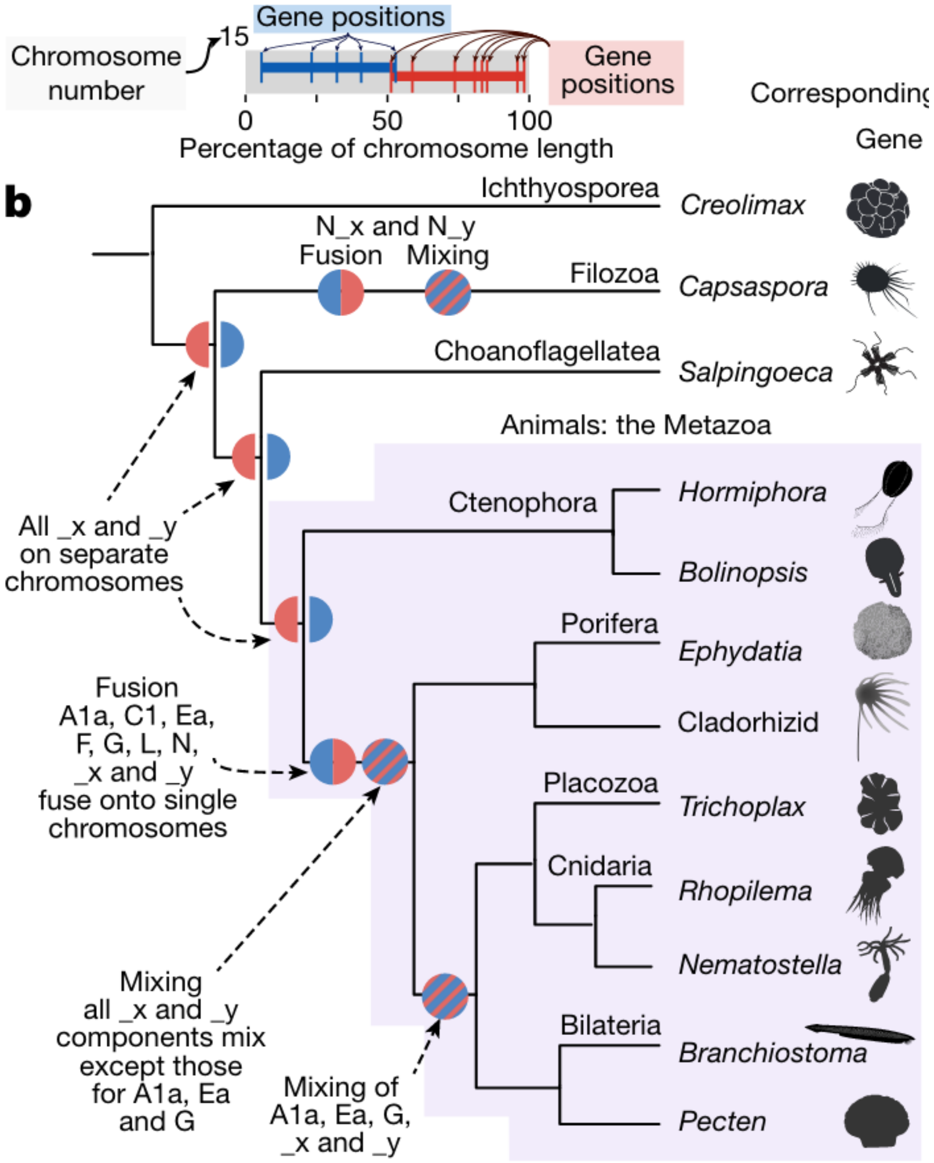
Jason Dobry (La Trobe University, Melbourne, Australia)

wikipedia.commons

Ancestral reconstruction of karyotype

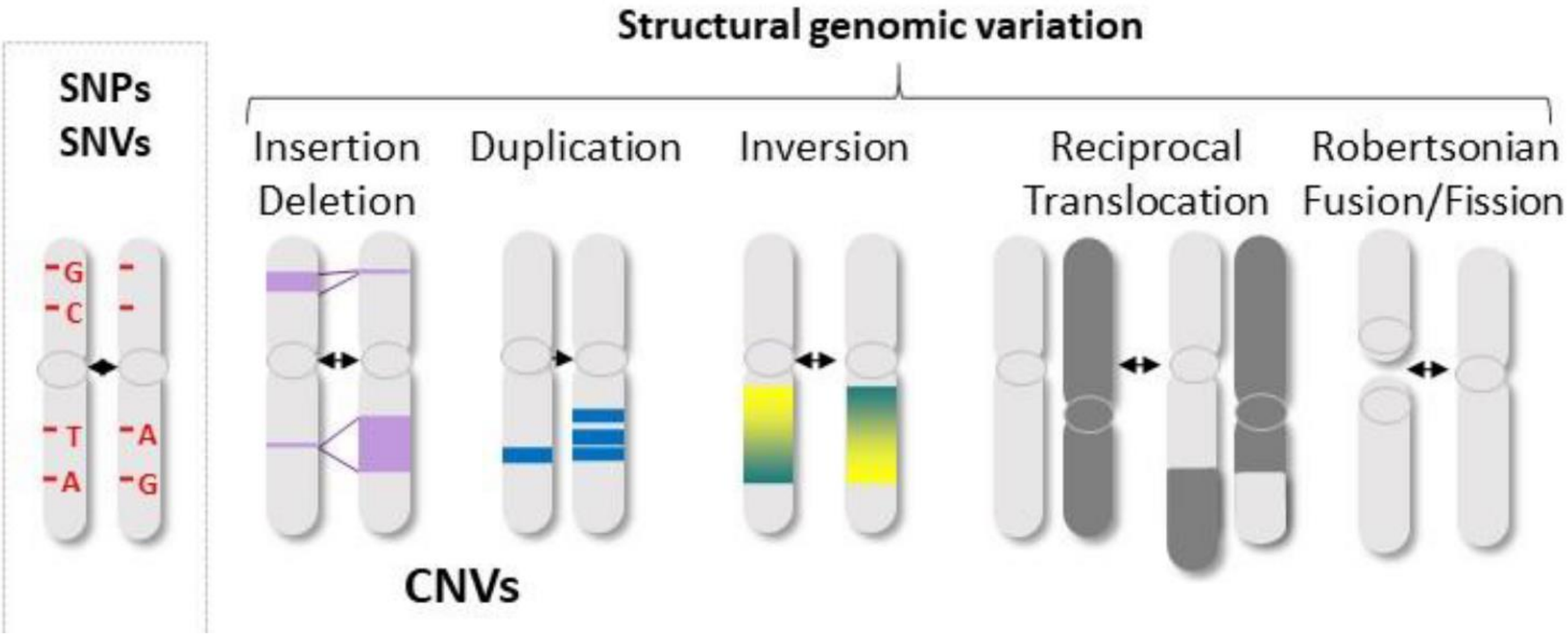
- Infer of rates and patterns of evolutionary change through time
- Resolve phylogenetic relationships
  - Chromosome fusion and subsequent mixing of gene order is an irreversible state
  - Ctenophora likely basal to all other Metazoa





Schultz et al. 2023

Genome structure – structural variants

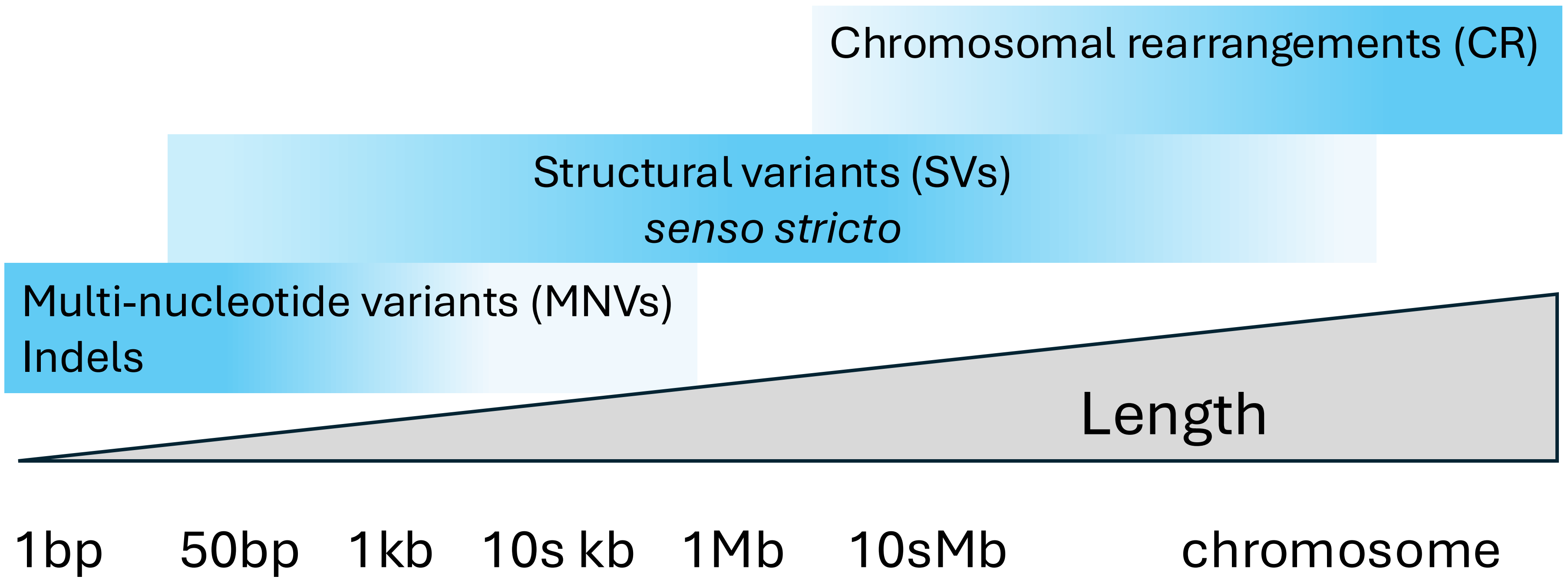


Single nucleotide variation

Copy number variation

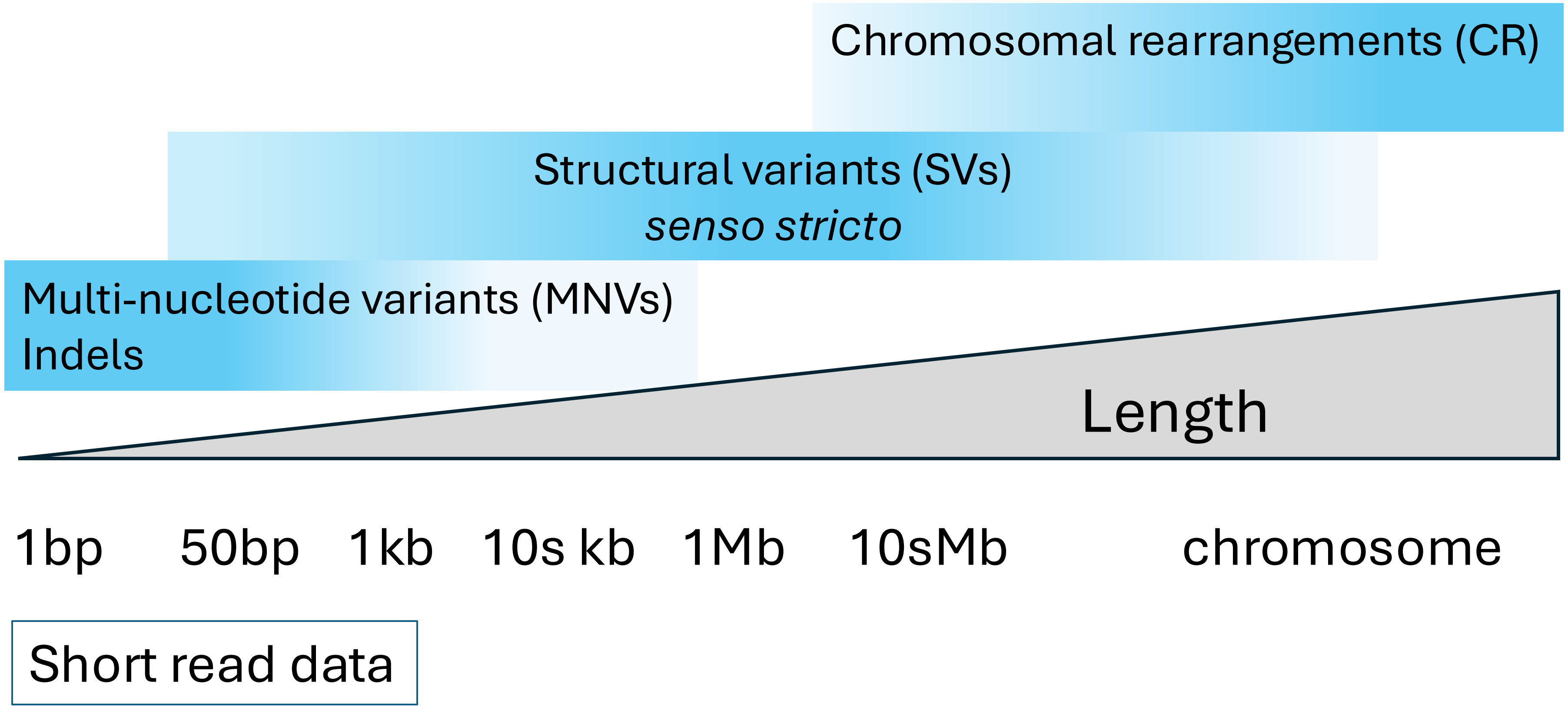
Berdan et al 2023

Genome structure – structural variants



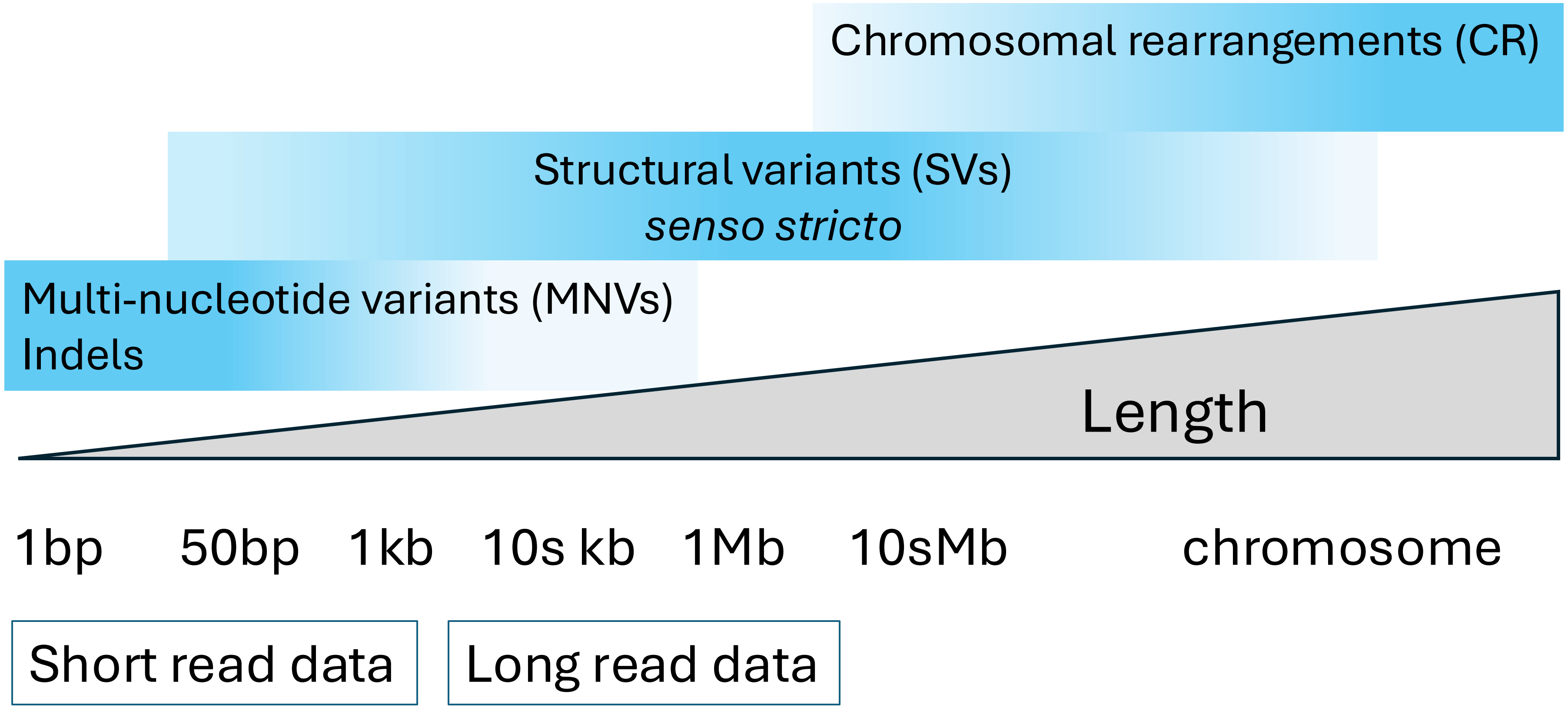
Berdan et al 2023

Genome structure – structural variants



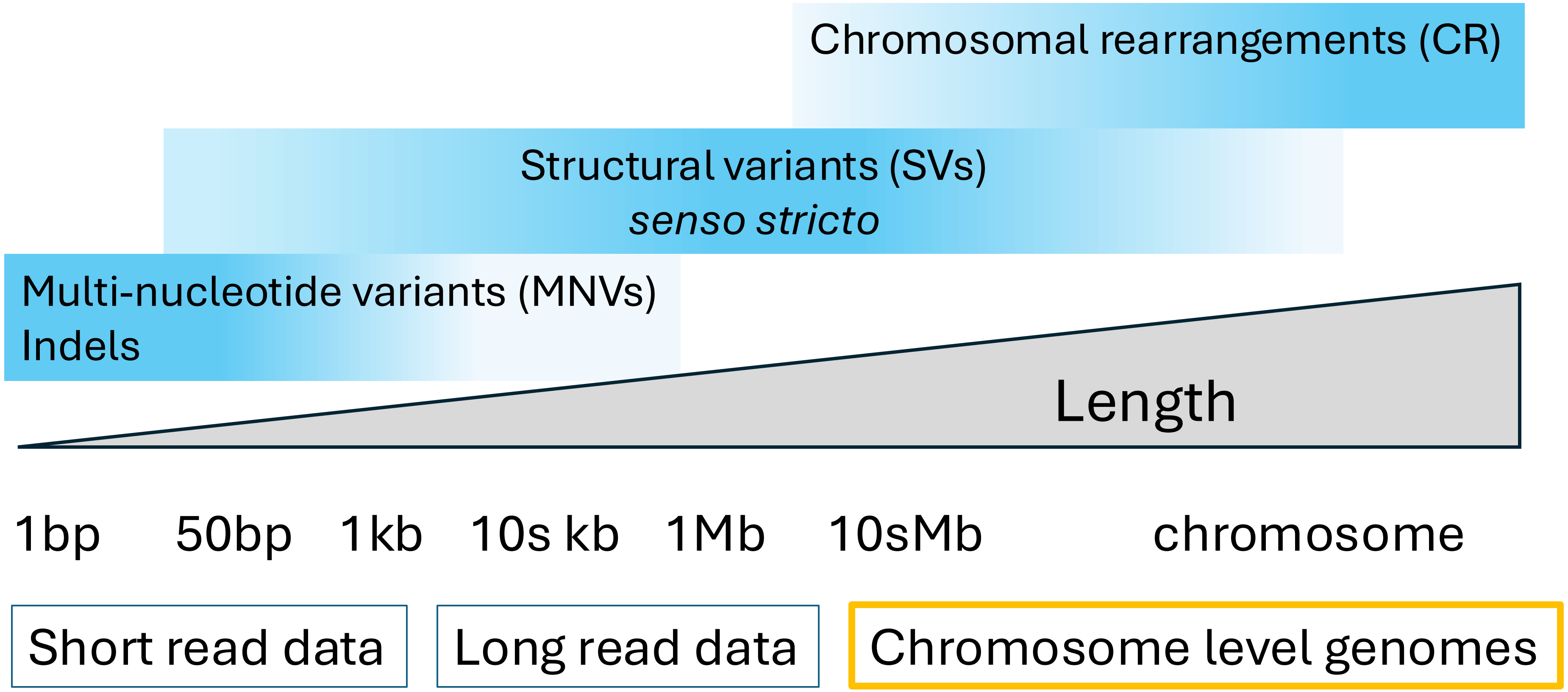
Berdan et al 2023

Genome structure – structural variants



Berdan et al 2023

Genome structure – structural variants



Berdan et al 2023

Genomic features

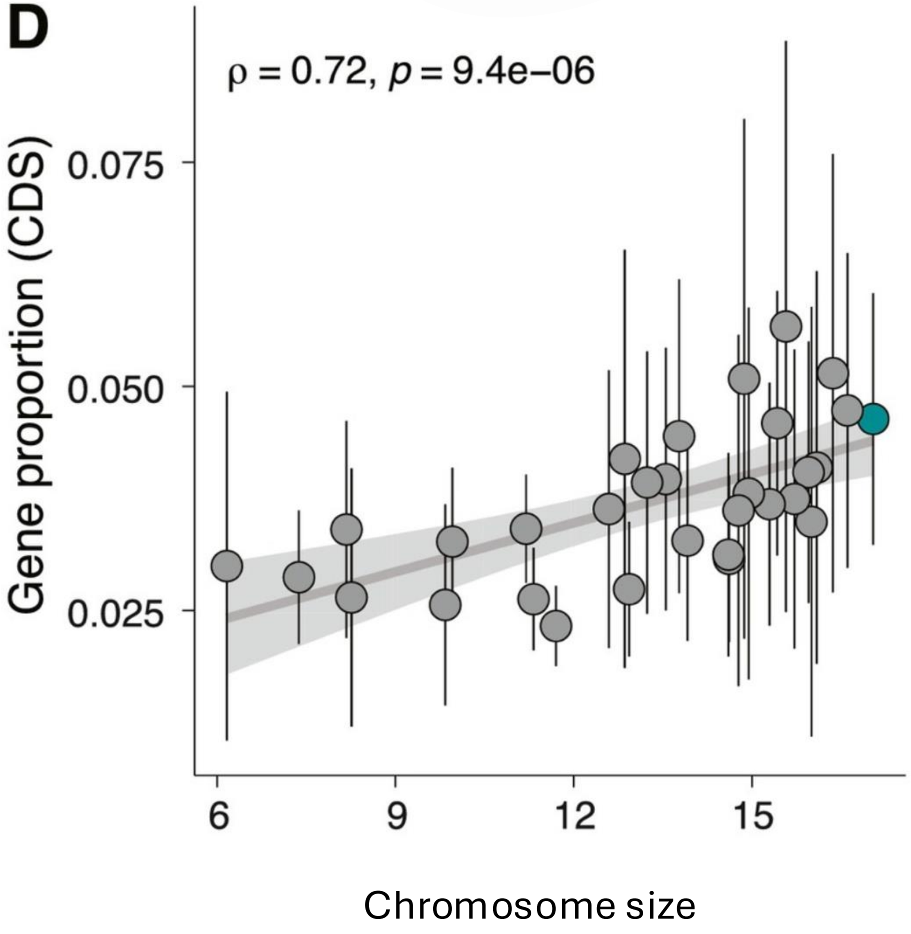
- Gene family expansion and contraction
- Regulatory element evolution
- Transposable element dynamics
- Genome composition evolution

Gene content

- protein coding genes and non-coding RNA within the genome
- Content vary across
  - genome
  - individuals
  - across species

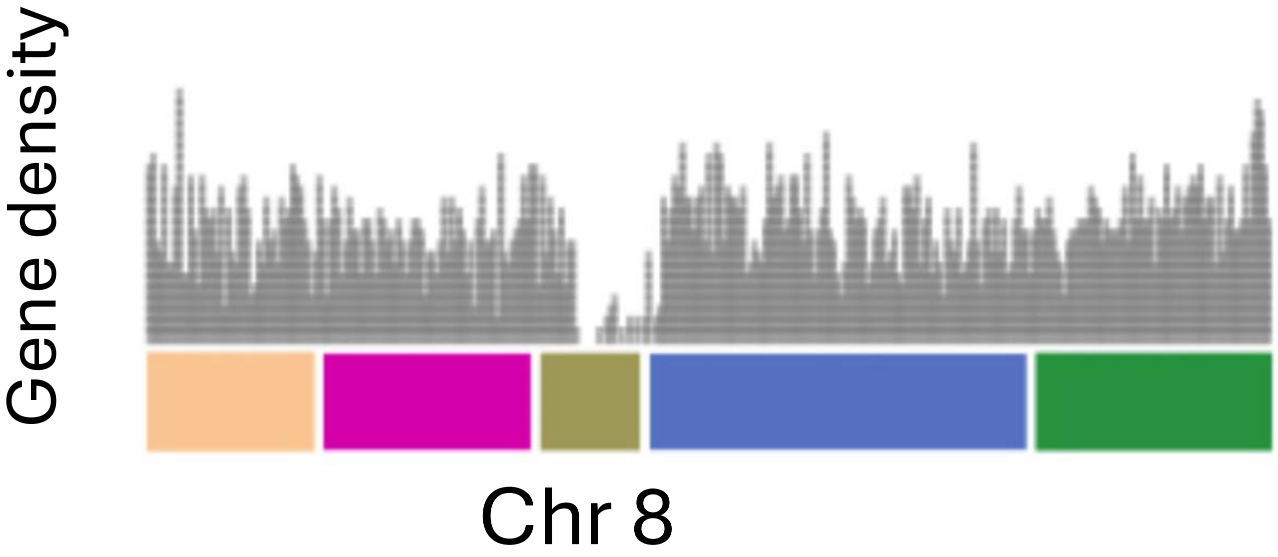
Chr 8

Arachis duranensis
Ren et al 2018

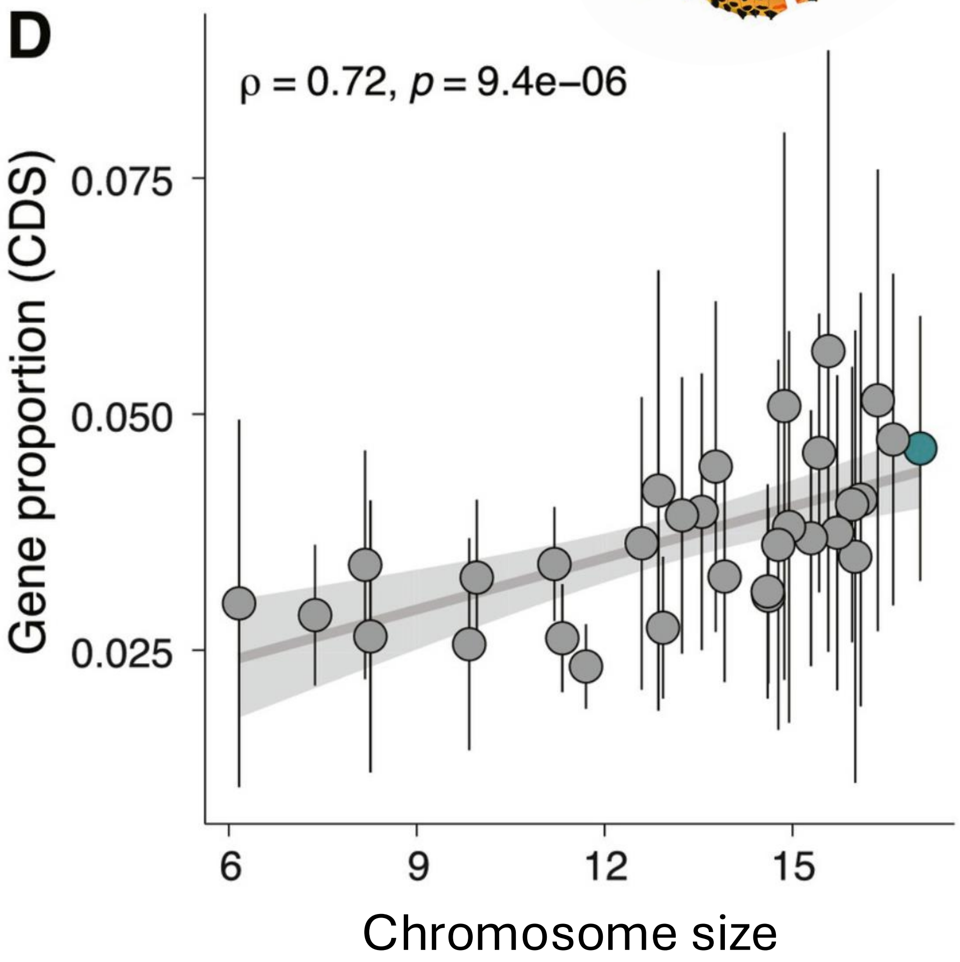


Shiplina et al 2022

Gene content across the genome



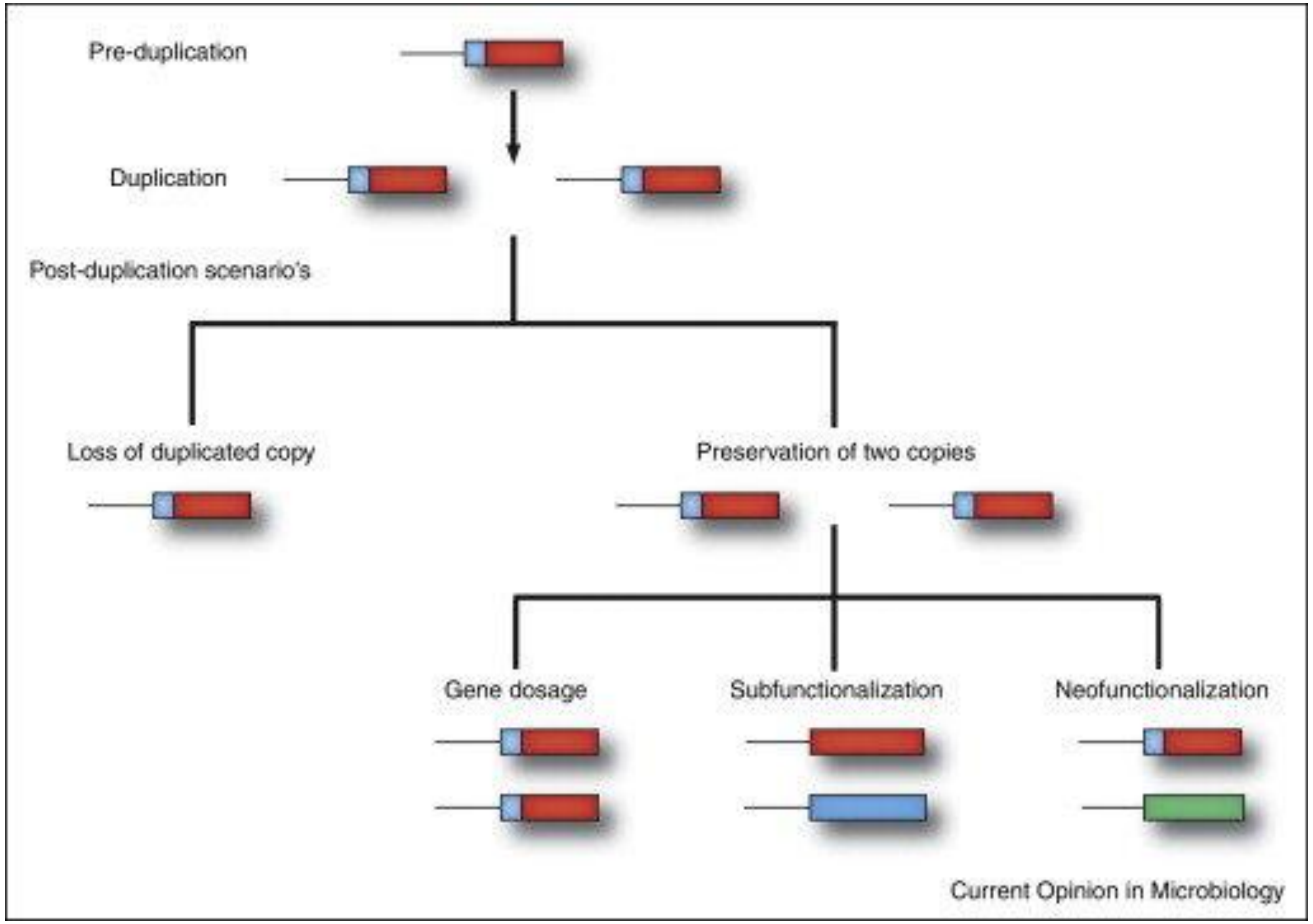
Arachis duranensis
Ren et al 2018



Vanessa cardui
Shiplina et al 2022

Gene duplications or deletions

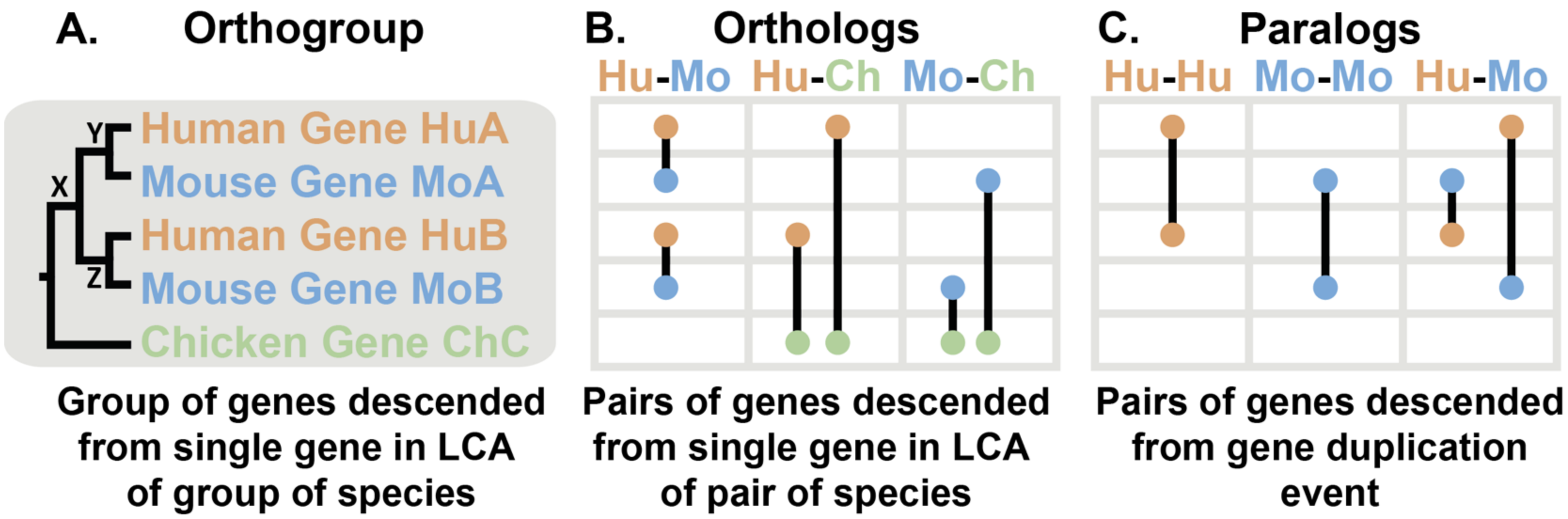
- Gene gain or loss leads to structural variation between individuals



Voordeckers & Verstrepen 2015

Gene family expansion or contraction

- Duplication of genes leads to differences between species
- Orthologs
- Paralogs

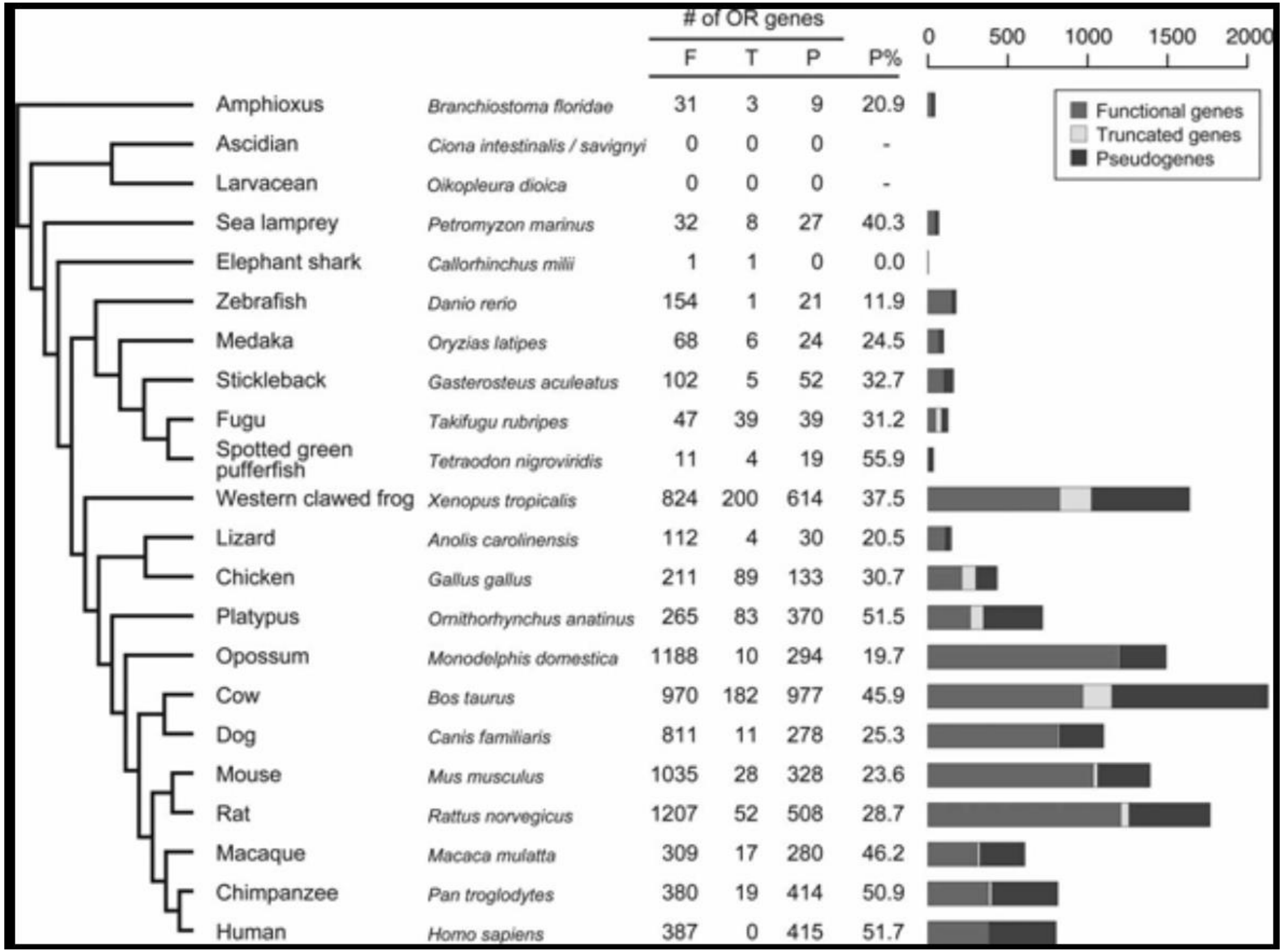


Emms and Kelly (2019)

Gene family expansion or contraction

- Expansion of the olfactory receptor (OR) gene family
- Large variation
  - 1,200 functional copies in rats
  - 400 in humans
  - 15 in pufferfish

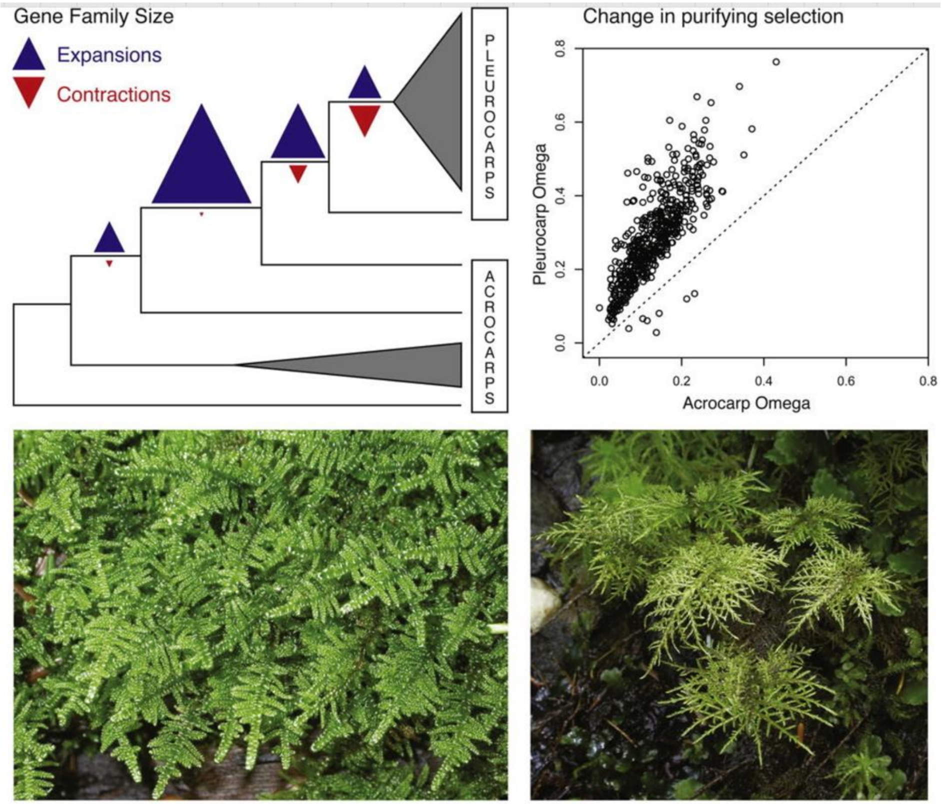
(Nimura, Hum Gen, 2009)



		F	T	P	P%
Amphioxus	*Branchiostoma floridae*	31	3	9	20.9
Ascidian	*Ciona intestinalis / savignyi*	0	0	0	-
Larvacean	*Oikopleura dioica*	0	0	0	-
Sea lamprey	*Petromyzon marinus*	32	8	27	40.3
Elephant shark	*Callorhinchus milii*	1	1	0	0.0
Zebrafish	*Danio rerio*	154	1	21	11.9
Medaka	*Oryzias latipes*	68	6	24	24.5
Stickleback	*Gasterosteus aculeatus*	102	5	52	32.7
Fugu	*Takifugu rubripes*	47	39	39	31.2
Spotted green pufferfish	*Tetraodon nigroviridis*	11	4	19	55.9
Western clawed frog	*Xenopus tropicalis*	824	200	614	37.5
Lizard	*Anolis carolinensis*	112	4	30	20.5
Chicken	*Gallus gallus*	211	89	133	30.7
Platypus	*Ornithorhynchus anatinus*	265	83	370	51.5
Opossum	*Monodelphis domestica*	1188	10	294	19.7
Cow	*Bos taurus*	970	182	977	45.9
Dog	*Canis familiaris*	811	11	278	25.3
Mouse	*Mus musculus*	1035	28	328	23.6
Rat	*Rattus norvegicus*	1207	52	508	28.7
Macaque	*Macaca mulatta*	309	17	280	46.2
Chimpanzee	*Pan troglodytes*	380	19	414	50.9
Human	*Homo sapiens*	387	0	415	51.7

Gene family expansion or contraction

- Cluster genes in gene families by sequence homology
- Infer phylogenetically informed rates of gain and loss
  - Birth-death model for evolutionary inferences
    - Café, Badirate



Hypnum imponens (photo: B. Shaw)　*Hylocomium splendens* (photo: B. Shaw)

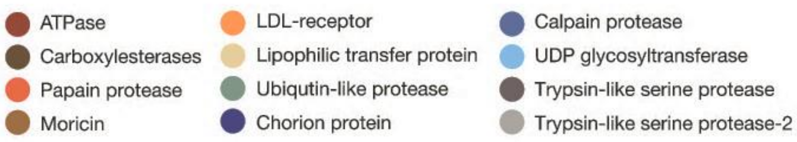
Johnson et al 2016

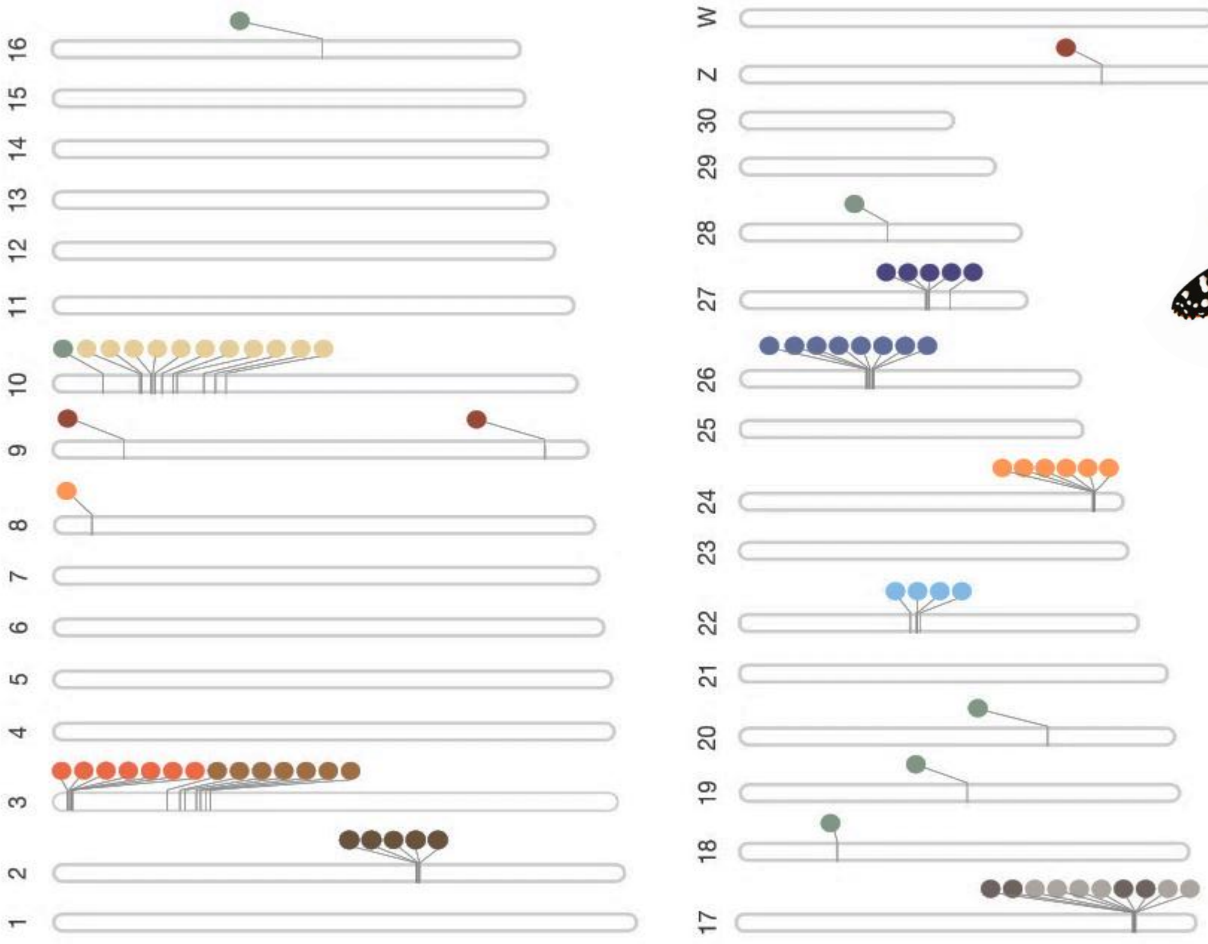
Gene family expansion or contraction

- Gene ontology analysis, approximation of gene function
  - Panther, TopGO
  - Domain analysis

- Functional validation
  - Gene editing/knockout

Candidate gene families





Shiplina et al 2022

Regulatory elements

- Promoters binding site for polymerase
- Proximal and distal enhancers, silencers, insulators

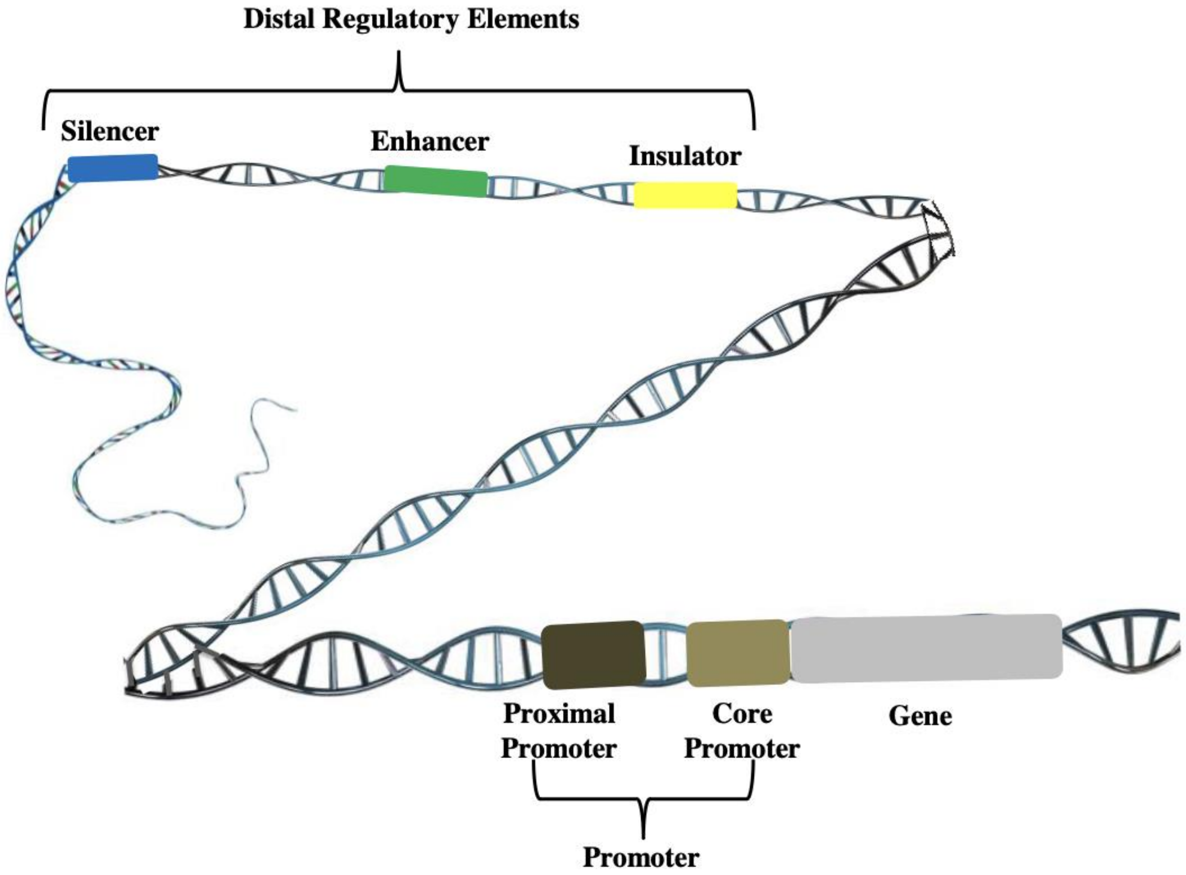
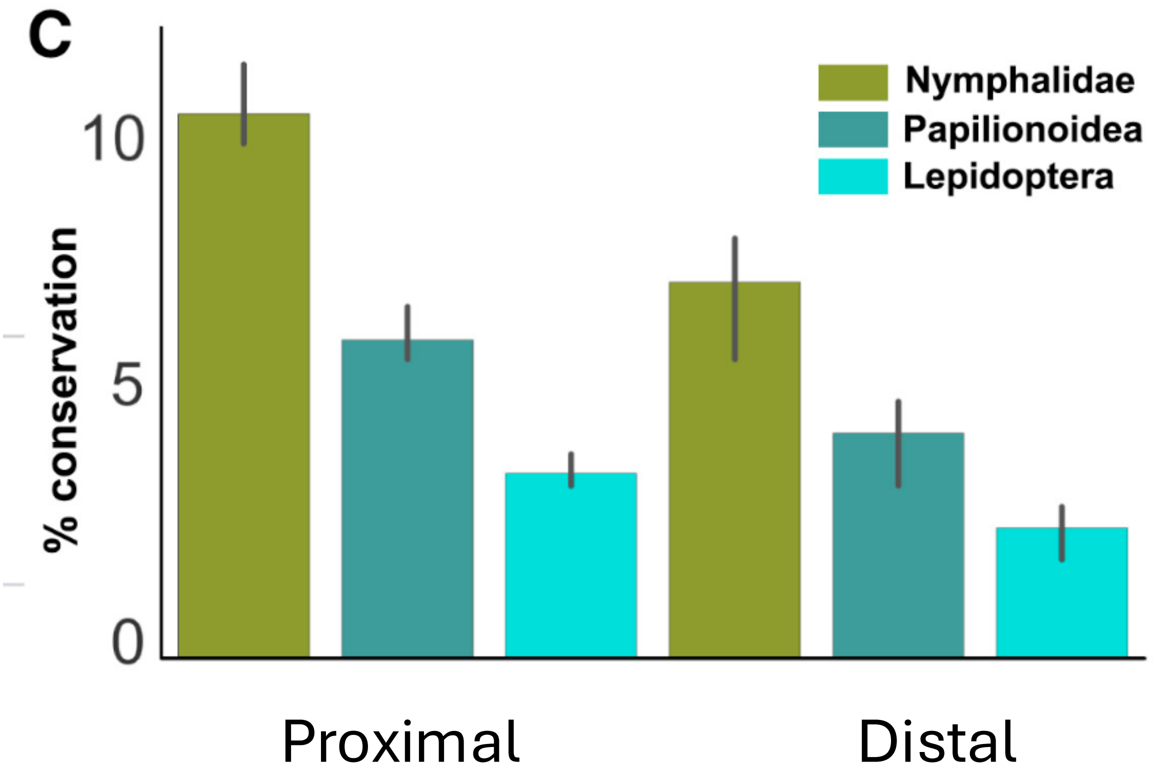


Fig. 1: Schematic representation of typical gene regulatory region.

Jadav et al 2016

Regulatory elements

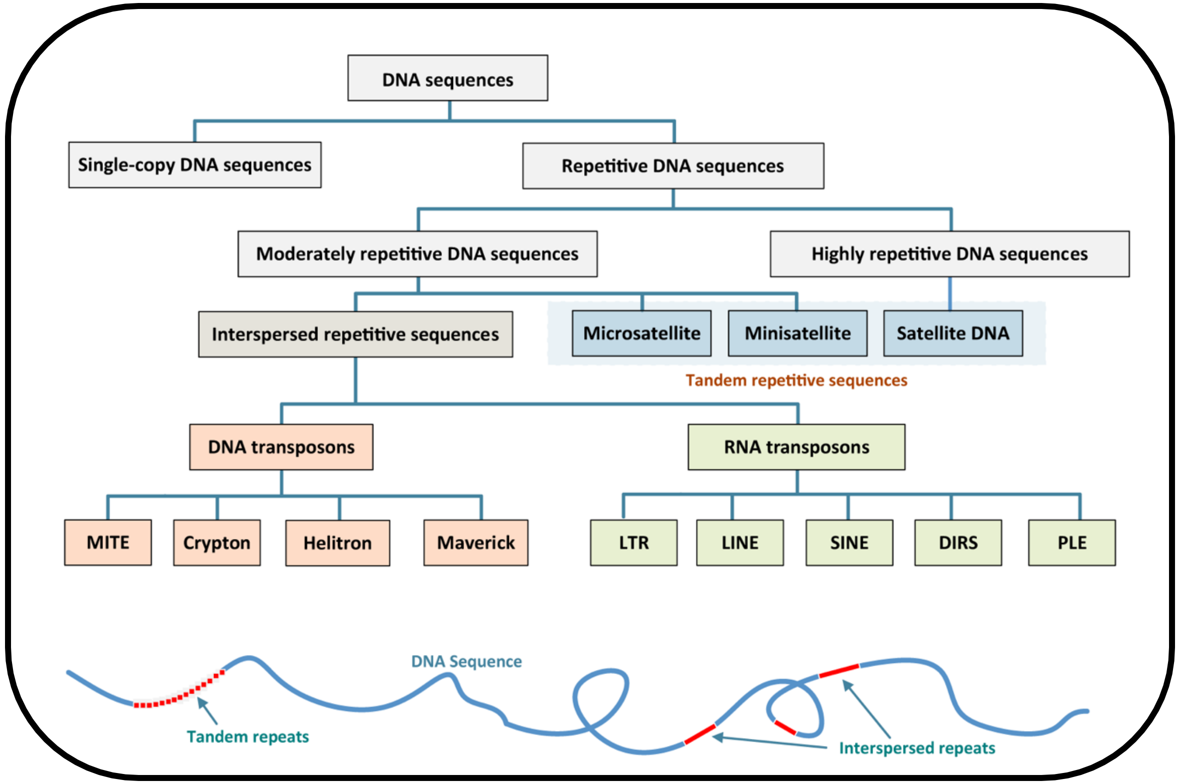
- Single nucleotide mutations, copy number variation (CNV) or TE insertions can change the regulatory function
- Infer by sequence homology
  - Whole genome alignment – conserved non-coding elements
  - Database search, known elements
- Prediction tools, specific motif
- Predict promoter - enhancer interactions in HiC-data
  - PSYCHIC (Ron et al 2017)
- Experimental approaches
  - ChIP-seq
  - ATAC-seq



Lewis et al. 2016

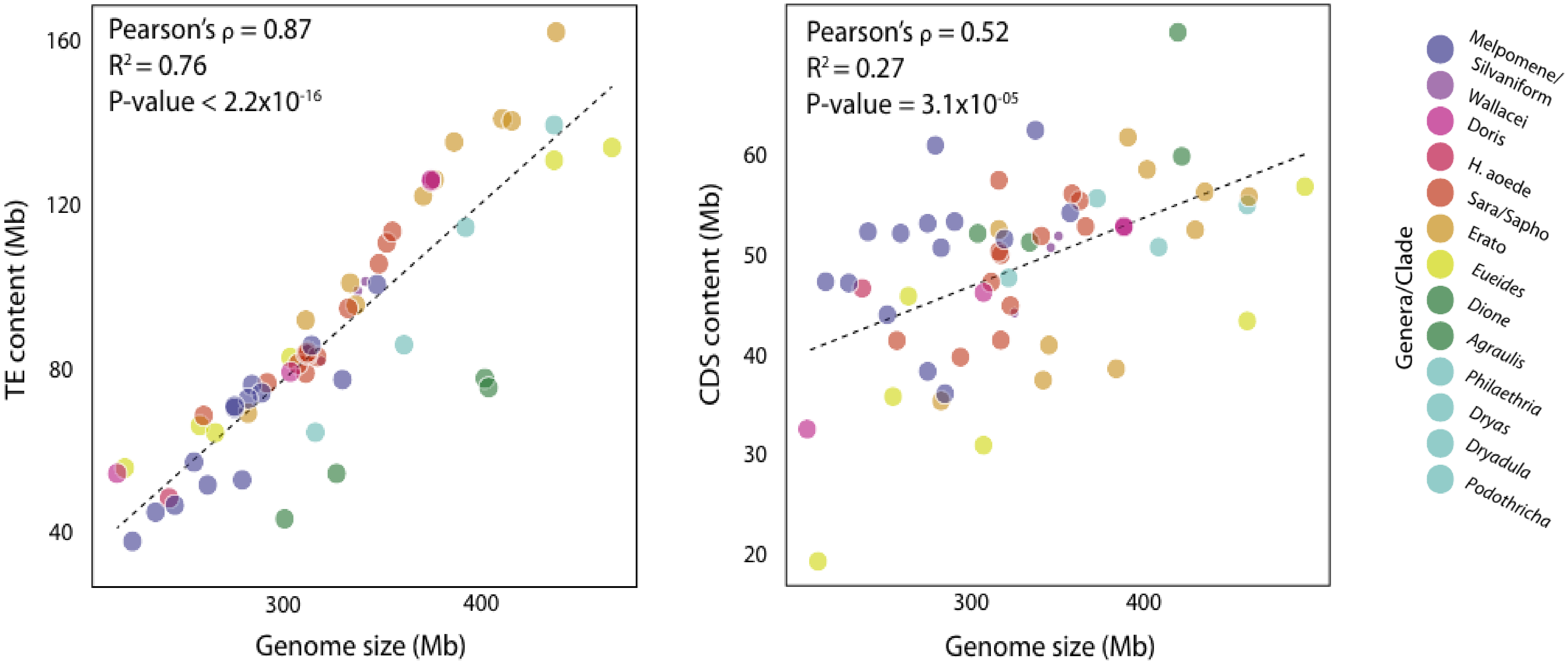
Repetitive sequences

- Transposable elements (mobile elements)
  - Move (or duplicate) from one location in the genome to another
  - Initiates their own transcription, polymerase or retrotranscription and insertion using a wide range of endonucleases
- Simple repeats
  - satellite repeats
  - microsatellites
- Multi-copy RNA genes
  - rRNA, tRNA, snRNA



Liao et al 2023

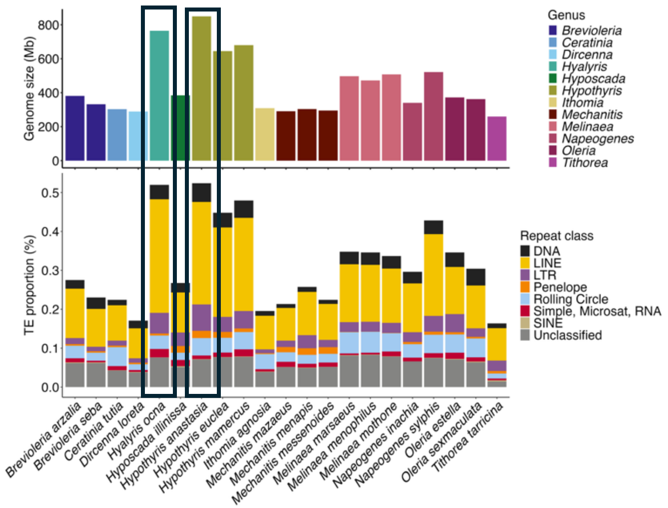
Transposable elements – genome size

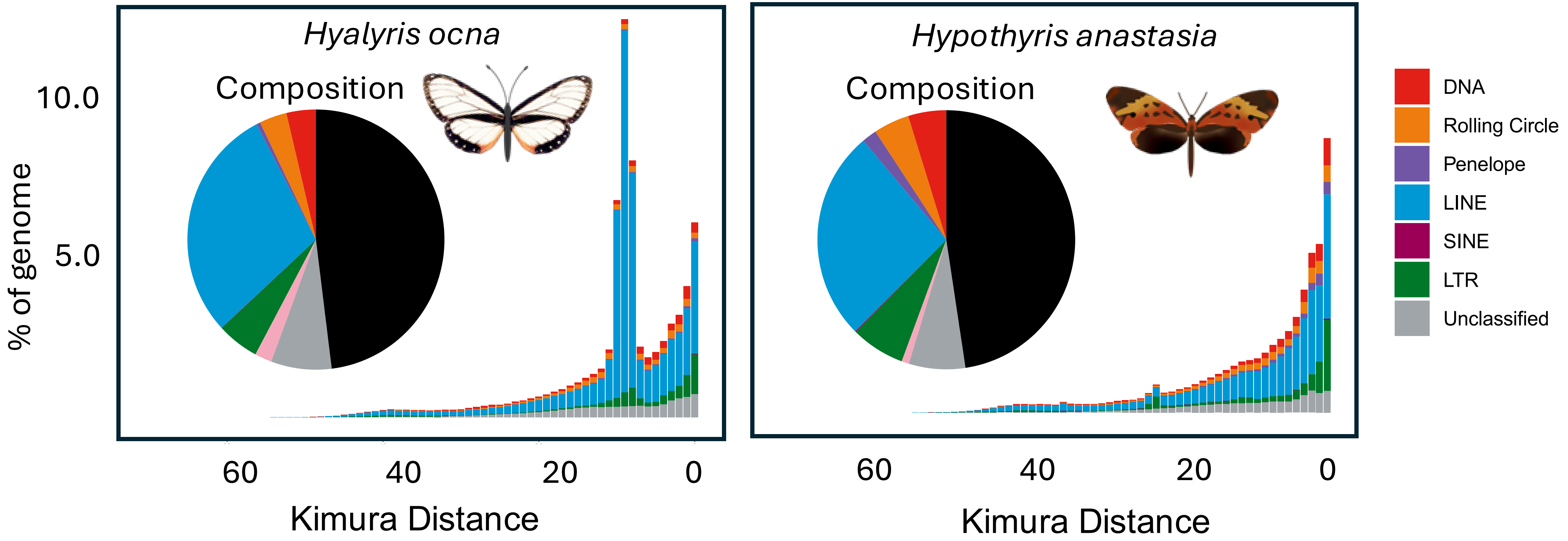


Ciccionardi et al. 2023

Dynamics through time

- Infer TE divergence time





Transposable elements - annotation

- TE prediction algorithms using structure and sequence homology
- Consensus building and classification steps
  - RepeatModeler2 (includes the predictors above)
- Manual curation

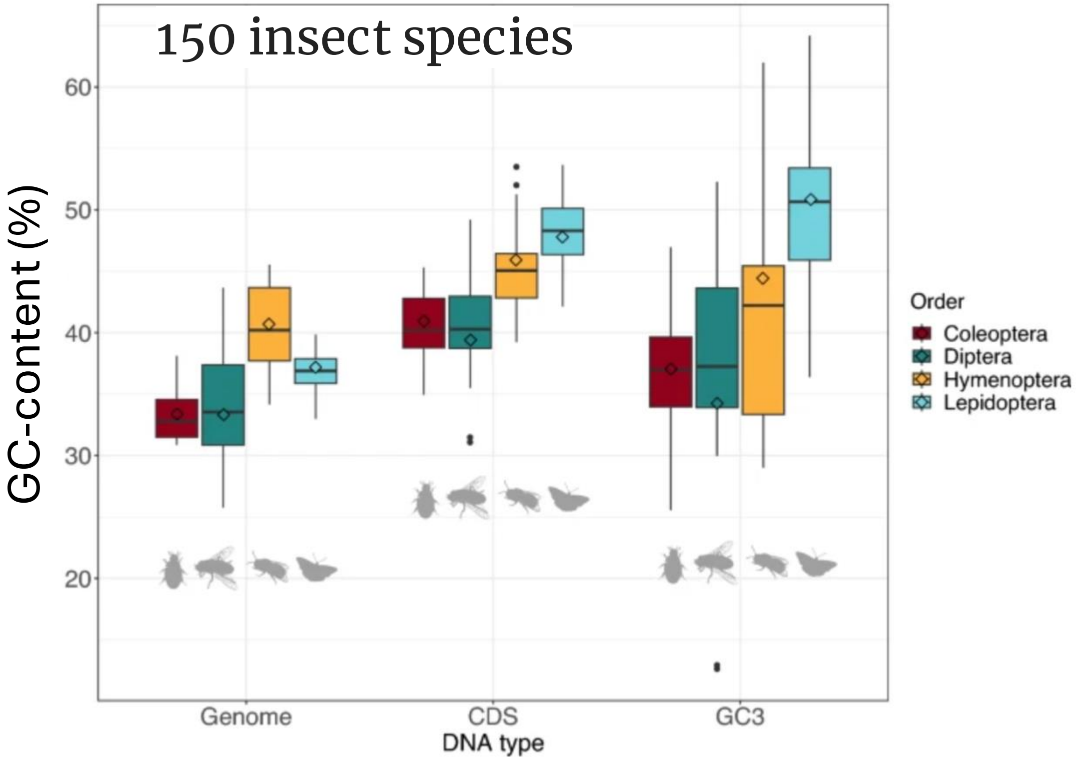
- EDTA, EarlGrey, Pantera

Satellite repeats

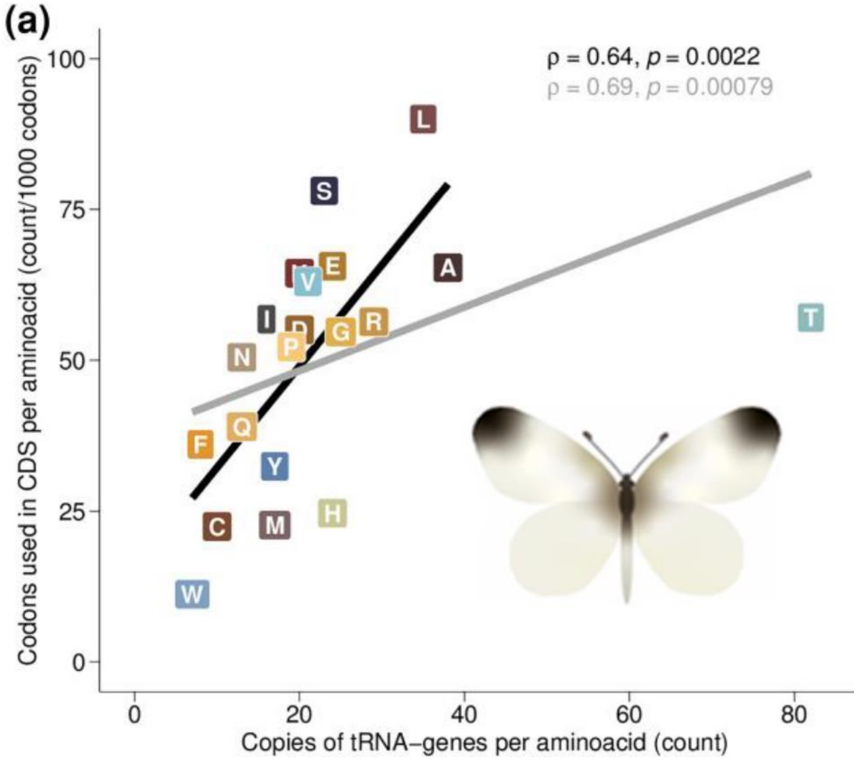
- Centromere-associated repeats
  - Associated tandem repeat arrays
  - Annotate location of centromere
- Telomeric repeats
  - TeloBase https://doi.org/10.1093/nar/gkad672
  - $(T_xA_yG_z)_n$
  - Ex, many insects $(TTAGG)_n$ $(TTCGGG)_n$
  - Detect differences in length or sequence shift
- Microsatellites

```
Ind1 ATATATATATAT
Ind1 ATATATATAT—
Ind2 ATATATATAT--
Ind2 --ATATATAT--
```

GC-content, codon usage and tRNA dynamics



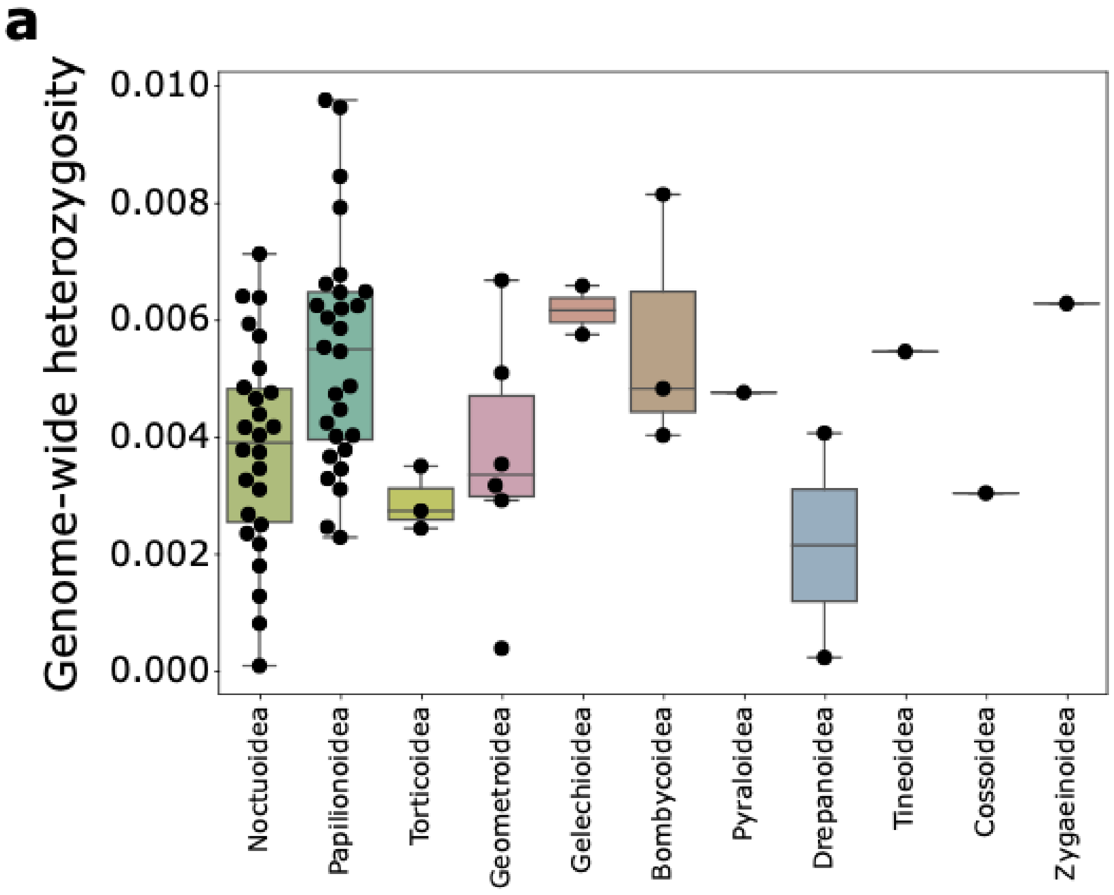
Kyriacou et al. 2024



Näsvall et al. 2023

Heterozygosity

- Number of segregating sites between the haplotypes
- Runs of homozygosity
  - Long uninterupted homozygous regions indicate low number of haplotypes segregating in the population
- Caveat: only one individual



Bortoluzzi et al 2023 bioRxiv

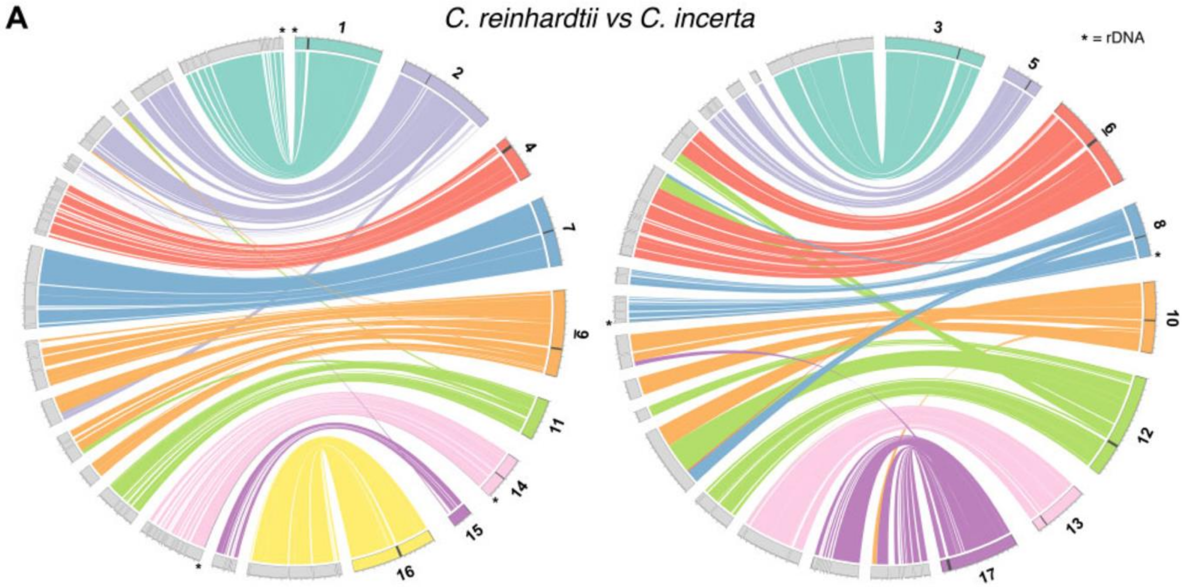
My work

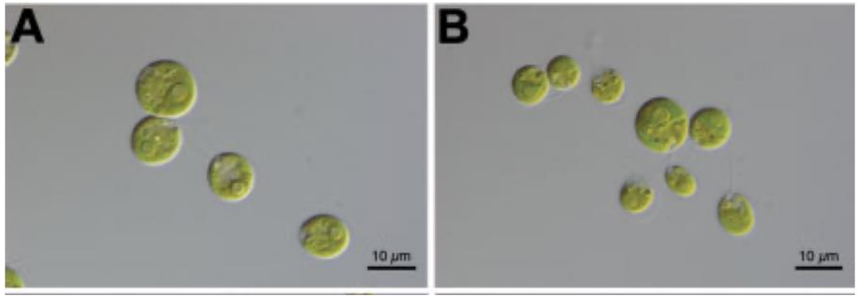
Synteny analysis - exercise

- Whole genome alignment
- Orthologous markers

Whole genome alignment

- Closely related taxa
  - MUMMER
  - Minimap2
  - Cactus
  - LastZ
  - D-Genies



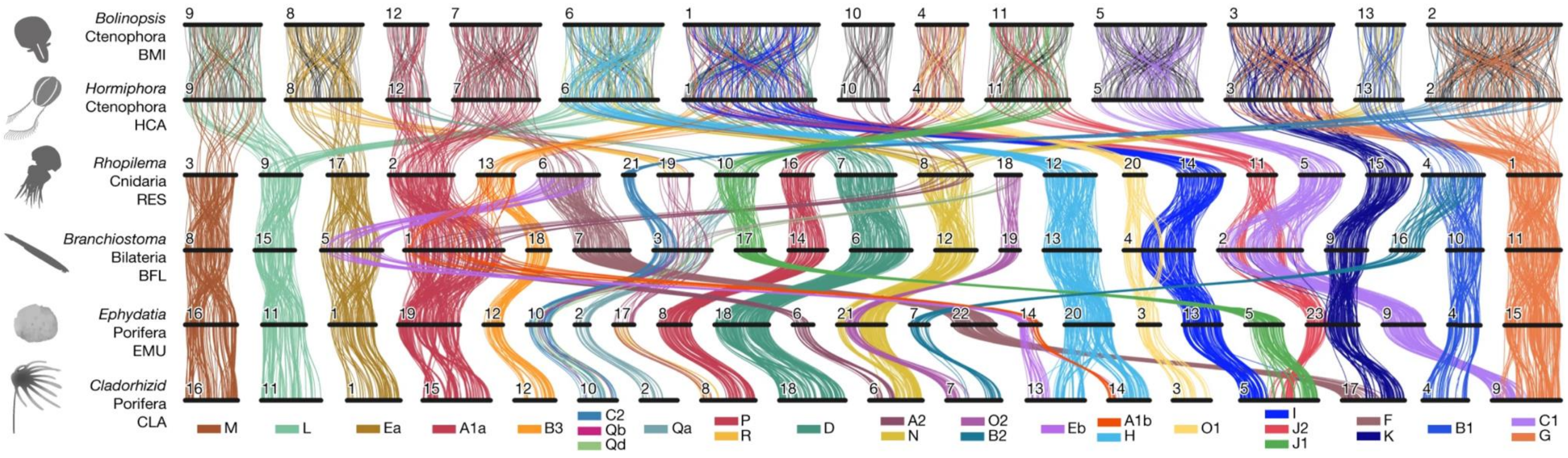


Green algae Chlamydomonas

Craig et al 2012

Orthologous markers

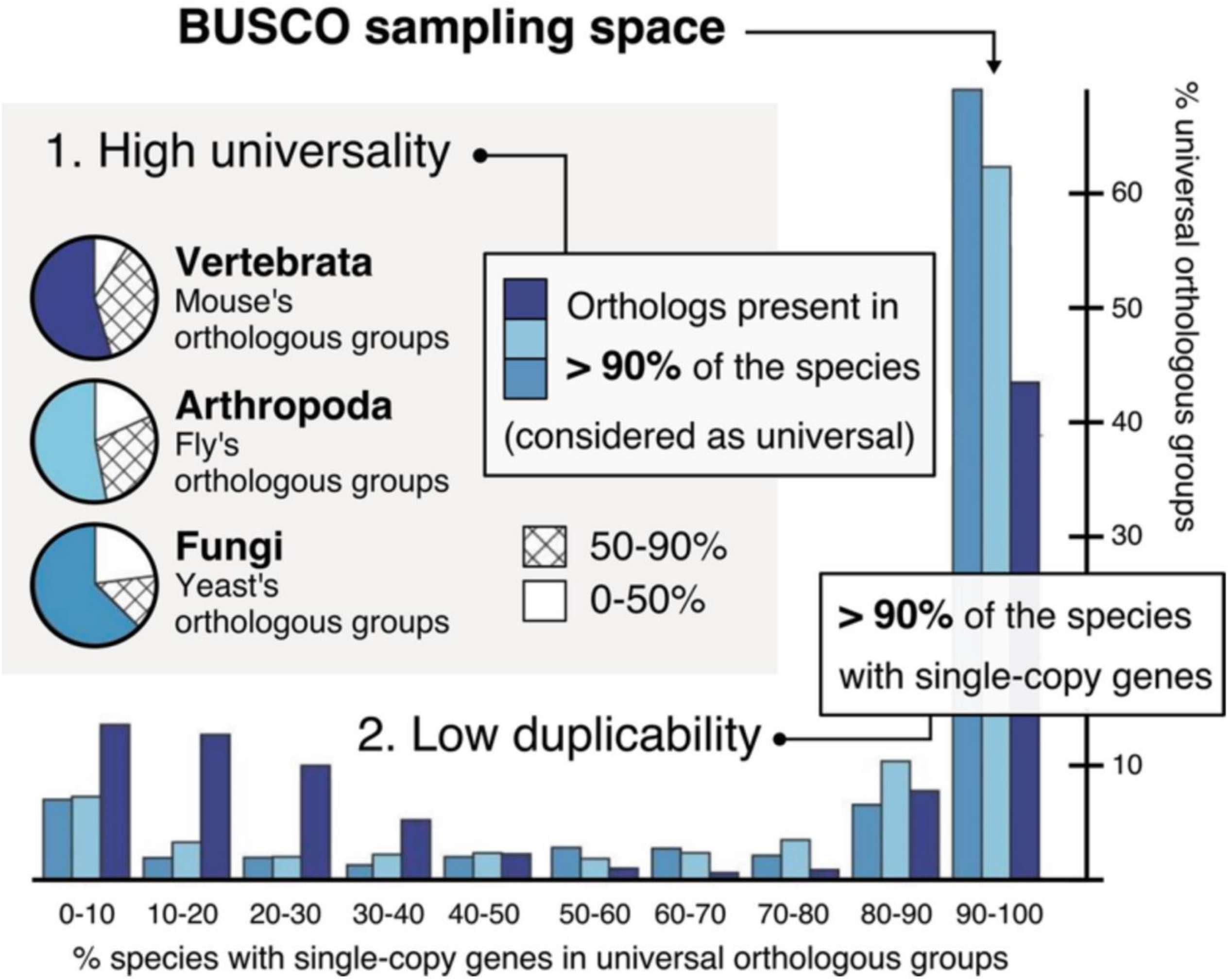
- Use conserved sequences as anchor markers
- Also for very divergent taxa



Schultz et al. 2023

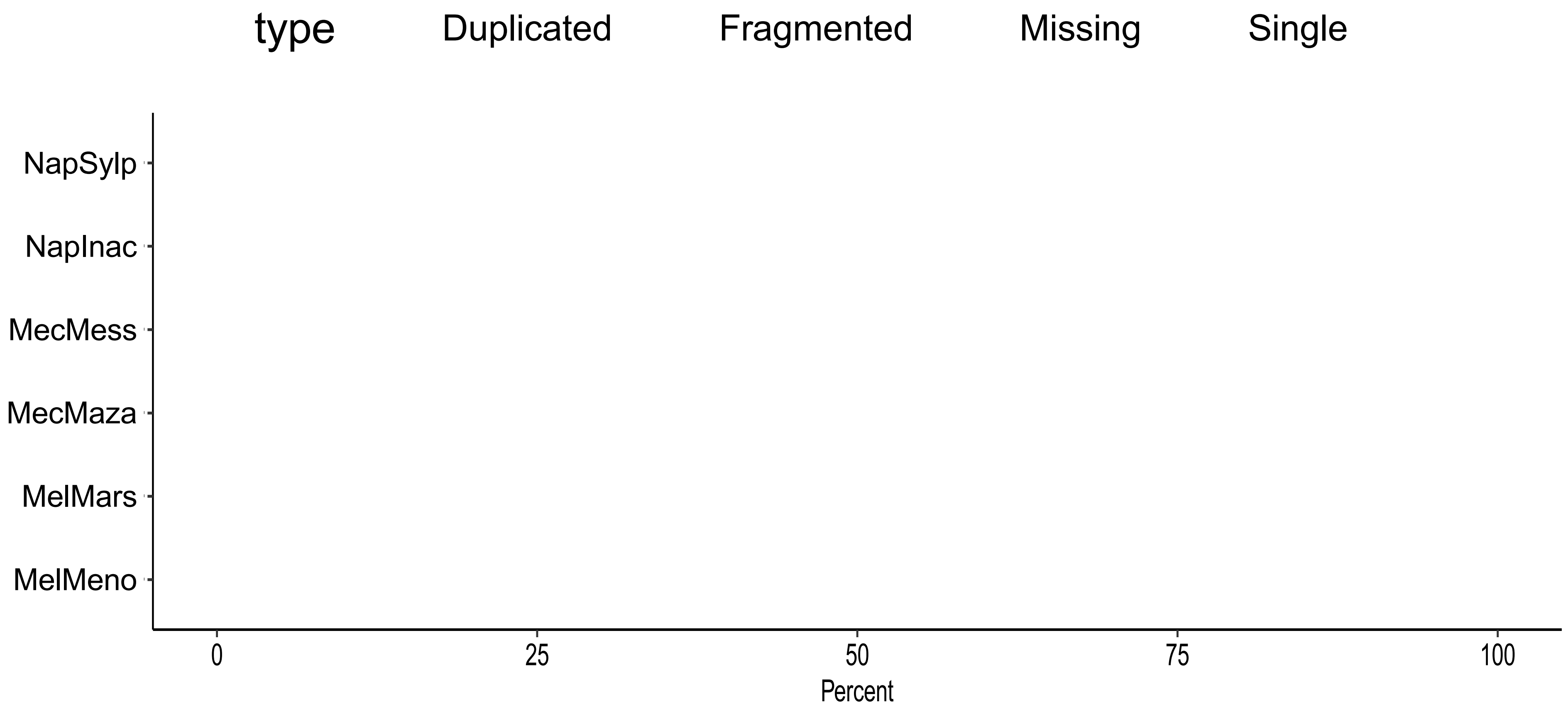
Orthologous markers

- BUSCO (Benchmarking Universal Single Copy Orthologues)
  - BLAST https://blast.ncbi.nlm.nih.gov/Blast.cgi
  - Finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance.



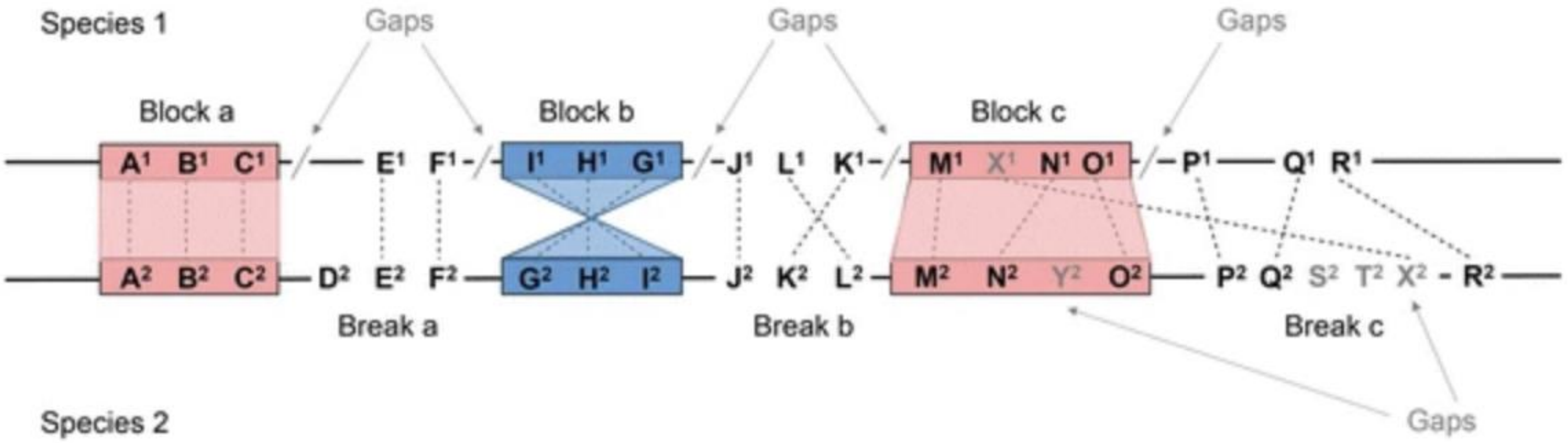
https://busco.ezlab.org

BUSCO completeness



Visualisation

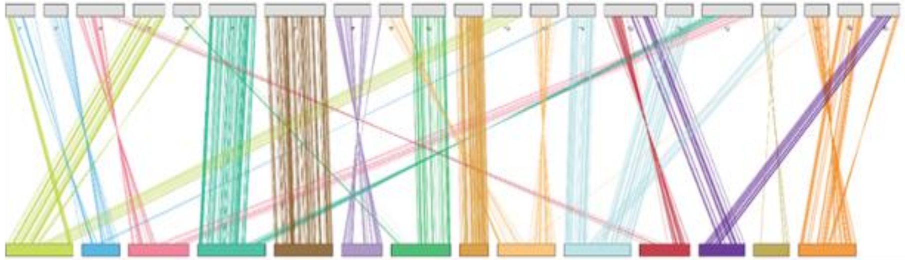
- Synteny



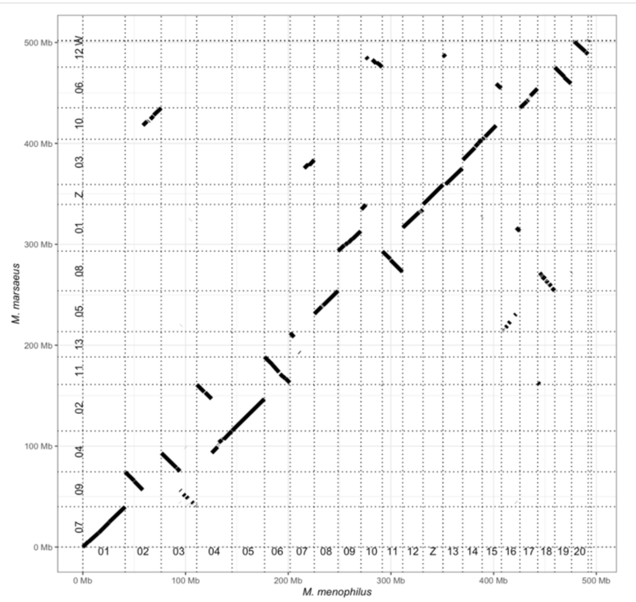
Visualisation

Melinaea marsaeus (n=14) vs *M. menophilus* (n=21)

Ribbon plot

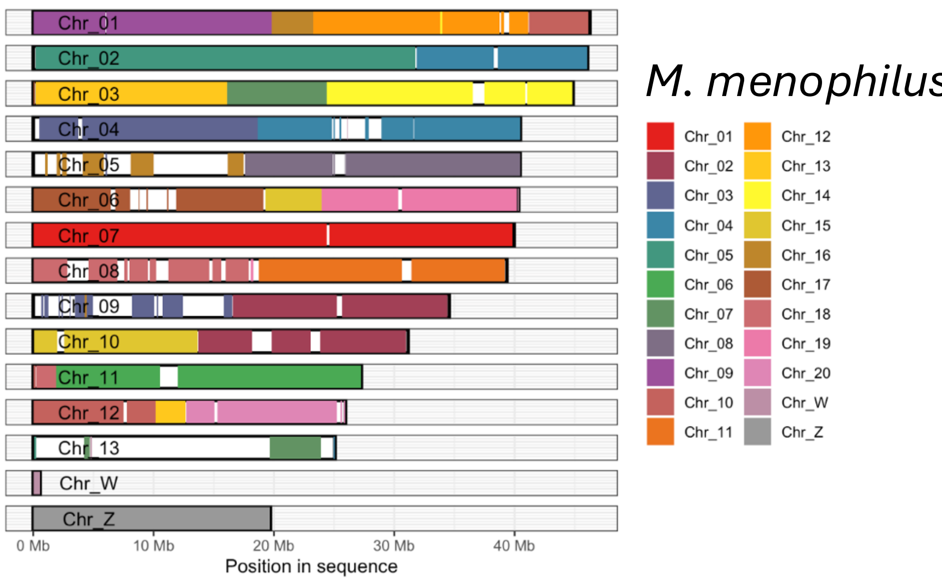


Dotplot



Chromosome painting

M. marsaeus



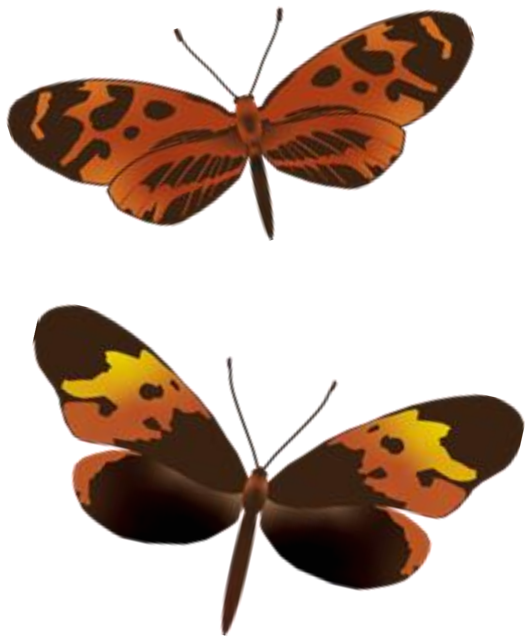
Hands-on synteny analysis

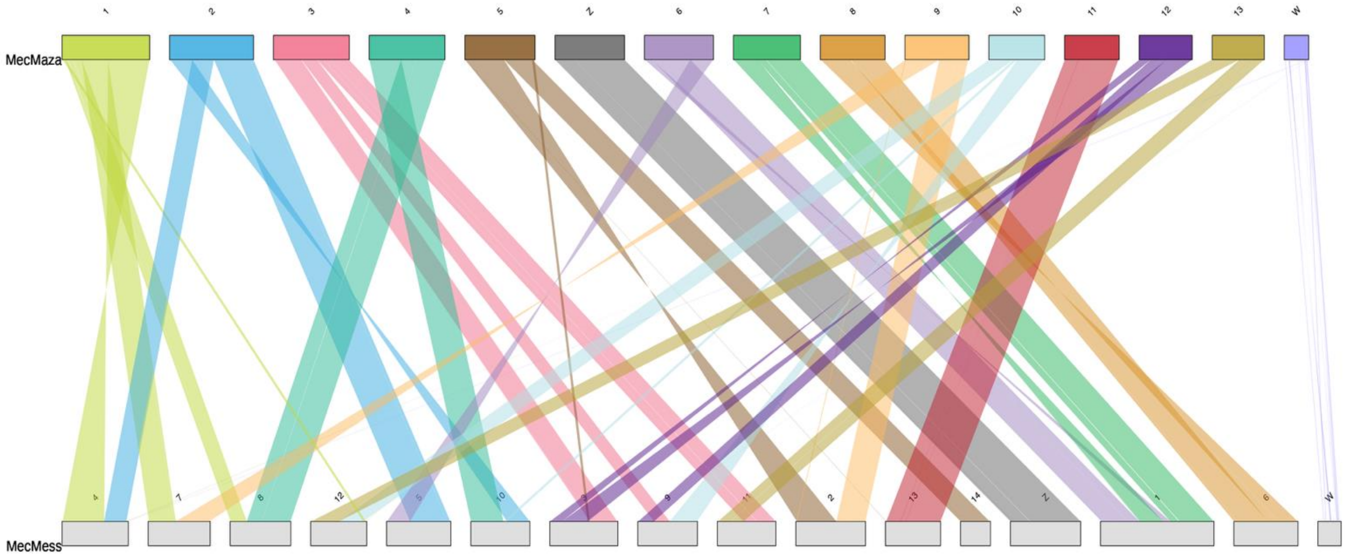
- Two species of Ithomiini
  - *Mechanitis marsaeus*
  - *Mechanitis messenoides*

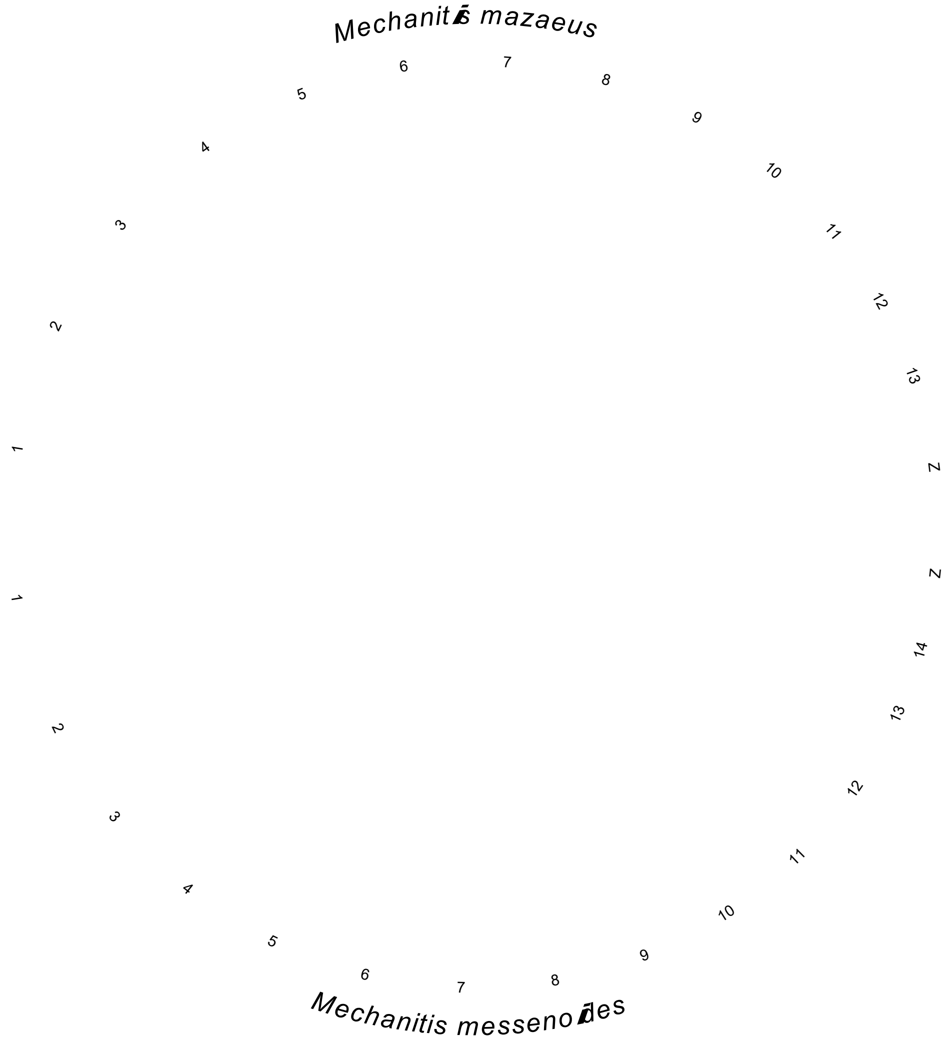
- Data type available
  - Reference genomes at NCBI

Hands-on synteny analysis

- Question:
  - Are there any chromosomal rearrangementsbetween the taxa?
- Methods
  - Whole genome alignment
    - MiniMap2
    - Ribbon plot
  - Marker orthology
    - BUSCO
    - Orthofinder
    - Circos
  - Extra: dotplot https://dgenies.toulouse.inra.fr




MecMaza
MecMess


Mechanitis mazaeus
Mechanitis messenoides